

5G RAN

Power Control Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between power reduction for CCUs and L/NR Inter-band Deep Coordination. For details, see 4.3.2.3 Function Impacts .	None	NR TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between the maximum SSB power offset and L/NR Inter-band Deep Coordination. For details, see 4.3.2.3 Function Impacts .	None	NR TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010010	Power Control	4 Downlink Power Control 5 Uplink Power Control

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Downlink power control	SRB power aggregation is supported only in SA networking. Inter-CC power aggregation is supported only in SA networking. None for other downlink power control functions	4 Downlink Power Control
Uplink power control	Simplified Msg4 power control configuration during random access is supported only in SA networking. Signaling power control optimization during random access is supported only in SA networking. None for other uplink power control functions	5 Uplink Power Control

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Downlink power control	Supported in both low and high frequency bands, with the following differences: PDCCH dynamic power control is supported only in low frequency bands. PDSCH dynamic power control is supported only in low frequency bands. PDSCH static power control is supported only when signaling is transmitted during RRC connection setup in high frequency bands. PCH static power control is supported only in high frequency bands. SRB power aggregation is supported only in high frequency bands. Inter-CC power aggregation is supported only in high frequency bands.	4 Downlink Power Control

Function Name	Difference	Chapter/Section
Uplink power control	Supported in both low and high frequency bands, with the following differences: SRS Po adaptation is supported only in low frequency bands. PUSCH joint power control is supported only in low frequency bands. Inter-CC PUSCH power control optimization is supported only in high frequency bands. Power control optimization for uplink CA is supported only in low frequency bands. Hybrid power control for uplink CA is supported only in high frequency bands.	5 Uplink Power Control

3 Overview

Due to path loss and shadow fading during radio propagation, the signal strength of electromagnetic waves received by a receiver from a transmitter decreases. To compensate for such impacts and enhance network coverage, the transmitter should increase its signal transmit power. However, if the transmitter uses excessively high transmit power, strong interference will be caused to other simultaneous transmissions, resulting in more energy being consumed. This is unfavorable to energy saving.

As such, **power control is required to achieve a balance among network coverage, interference suppression, and energy consumption by adjusting the transmit power of the gNodeB and UE.**

There are two types of power control in an NR system: uplink power control and downlink power control.

- Downlink power control adjusts the power of each downlink channel or signal by configuring an offset relative to the cell reference power. The total power of a single channel is evenly allocated to each resource element (RE) as downlink transmit power per RE, which is called cell reference power.
- Uplink power control adjusts the power of each uplink channel or signal for a UE by adjusting the transmit power control (TPC) value and considering information such as UE's maximum transmit power, the nominal power of the cell, the path loss, the modulation and coding scheme (MCS), and the amount of radio resources.

NOTE

The total power of a single channel is a fixed value determined by the module capability of the base station. In practice, the total power of a single channel can be configured based on the value of **MaxTransmitPower** as described in [Cell Reference Power Calculation](#) in [4.1.1 Static Power Control](#).

4 Downlink Power Control

Downlink power control includes both static and dynamic power control. Static power control adjusts transmit power based on the coverage capability of each channel or signal by configuring power offset parameters, while dynamic power control adaptively adjusts transmit power based on the actual transmission status and UE feedback of each channel or signal. Specifically,

- Static power control is applicable to the following downlink physical channels and signals:
 - Physical downlink shared channel (PDSCH) (only in high frequency bands)
 - Physical downlink control channel (PDCCH)
 - Physical broadcast channel (PBCH)
 - Synchronization signal (SS)
 - Tracking reference signal (TRS)
 - Channel state information-reference signal (CSI-RS)Static power control is applicable to the PDSCH only when signaling is transmitted during RRC connection setup in high frequency bands.
- Dynamic power control is applicable to the following downlink physical channels:
 - PDCCH
 - PDSCH

4.1 Principles

4.1.1 Static Power Control

Static power control for all downlink channels and signals is achieved by configuring offsets to the cell reference power (ReferencePwr). Only the power offset is specified by different parameters for each channel or signal.

The actual transmit power of downlink channels or signals on each RE is calculated using the following formula:

$$\text{ReferencePwr} + \text{PowerOffset} + 10 \times \log_{10}(\text{RFChannelTXNum})$$

where

- **ReferencePwr** is the cell reference power (in the unit of dBm). Its calculation is determined by **MaxTransmitPower** and **RBcell**, as explained in [Cell Reference Power Calculation](#).
- **PowerOffset** is the effective power offset of each channel or signal relative to the cell reference power. Its configuration is explained in [Power Offset Configuration](#).
- **RFChannelTXNum** is the number of physical RF transmit channels.

For example: In a 100 MHz 64T NR TDD cell with the subcarrier spacing of 30 kHz, if the maximum power offset of the PBCH and SS (**PowerOffset**) is 3 dB, **MaxTransmitPower** is 30.9 dBm, and **RBcell** is 273, then the cell reference power (**ReferencePwr**) is -4.25 dBm and the power of the PBCH and SS on each RE is 16.8 dBm. The same calculation rules apply to other channels or signals.

 NOTE

The cell reference power is the power of a single physical RF transmit channel on each RE.

Cell Reference Power Calculation

The cell reference power (**ReferencePwr**) is calculated using the following formula:

$$\text{ReferencePwr} = \text{MaxTransmitPower} - 10 \times \log_{10}(\text{RBcell} \times 12)$$

where

1. **MaxTransmitPower** is the maximum transmit power of each physical RF transmit channel (in the unit of dBm or mW). Its configuration is described in [Table 4-1](#).

Table 4-1 Configuration of the maximum transmit power of each physical RF transmit channel

RAT	Base Station Model	Configuration
Low-frequency TDD	3900 and 5900 series base stations (macro base stations)	MaxTransmitPower can be configured based on the unit of transmit power specified by the NRDUCellTrp.PowerConfigMode parameter. • It is specified by the NRDUCellTrp.MaxTransmitPower parameter when the above parameter is set to TRANSMIT_POWER. • It is specified by the NRDUCellTrp.MaxTransmitPowerMw parameter when the above parameter is set to TRANSMIT_POWER_MW.

RAT	Base Station Model	Configuration
	DBS3900 LampSite and DBS5900 LampSite	You are advised to use either of the following configuration methods based on a consistent unit of transmit power to facilitate operation and maintenance. MaxTransmitPower is specified by the NRDUCellTrp.MaxTransmitPower or NRDUCellCoverage.MaxTransmitPower parameter if the parameter specifying the power configuration mode for a cell transmission reception point (TRP)^a is set to TRANSMIT_POWER and the parameter specifying the power configuration mode for the cell coverage area^b is set to TRANSMIT_POWER_DBM. When both NRDUCellTrp.MaxTransmitPower and NRDUCellCoverage.MaxTransmitPower are configured, the latter takes effect preferentially^c. MaxTransmitPower is specified by the NRDUCellTrp.MaxTransmitPowerMw or NRDUCellCoverage.MaxTransmitPowerMw parameter if both the parameter specifying the power configuration mode for a cell TRP and the parameter specifying the power configuration mode for the cell coverage area are set to TRANSMIT_POWER_MW. When both NRDUCellTrp.MaxTransmitPowerMw and NRDUCellCoverage.MaxTransmitPowerMw are configured, the latter takes effect preferentially^c.
High - frequency TDD	3900 and 5900 series base stations (macro base stations)	$\text{MaxTransmitPower} = \text{EIRP}_{\text{Cell}}(\text{dBm}) - \text{Gain}(\text{dBi}) - 10 \times \lg(N_{\text{Tx}})$ where $\text{EIRP}_{\text{Cell}}(\text{dBm})$ is specified by the NRDUCellTrp.MaxEirp parameter. - Gain(dBi) is the antenna gain of the RF module serving a cell. For details about antenna gains, see the technical specifications in <i>3900 & 5900 Series Base Station Product Documentation</i>. N_{Tx} is the number of transmit antennas, which is specified by the NRDUCellTrp.TxRxMode parameter.

RAT	Base Station Model	Configuration
		a: The power configuration mode of a cell TRP is specified by the NRDUCellTrp.PowerConfigMode parameter. b: The power configuration mode of the cell coverage area is specified by the NRDUCellCoverage.PowerConfigMode parameter. c: The parameter for a cell TRP (NRDUCellTrp.MaxTransmitPower or NRDUCellTrp.MaxTransmitPowerMw) takes effect for a single TRP, while the parameter for the cell coverage area (NRDUCellCoverage.MaxTransmitPower or NRDUCellCoverage.MaxTransmitPowerMw) takes effect for the low-frequency RF module of the corresponding LampSite base station. The former is configured by default, while the latter is configured based on the actual power requirements of the low-frequency RF module. If both are configured, the latter takes effect preferentially.

2. **RBcell** is the number of resource blocks (RBs) corresponding to the total bandwidth of a cell. Each RB contains 12 REs.

Power Offset Configuration

PowerOffset is the effective power offset of each channel or signal relative to the cell reference power. The effective offset may be less than the maximum offset due to limited power aggregation capabilities of RF modules or insufficient power resources during scheduling. The maximum power offset of each channel or signal relative to the cell reference power can be configured as described in [Table 4-2](#).

Table 4-2 Configuration of the maximum power offset of each channel or signal

Channel or Signal	Configuration
PBCH and SS	• In NR TDD low-frequency cells, the maximum power offset of the PBCH and SS is specified by the NRDUCellTrpBeam.MaxSsbPwrOffset parameter. • In NR TDD high-frequency cells, the maximum power offset of the PBCH and SS is specified by the NRDUCellTrpMmwavBeam.MaxSsbPwrOffset parameter.
PDCCH	The maximum power offset of the PDCCH is specified by the NRDUCellChnPwr.MaxCommonDciPwrOffset parameter.
TRS	The maximum power offset of the TRS is specified by the NRDUCellChnPwr.TrsPwrOffset parameter.
Paging channel (PCH) carried on the PDSCH	The maximum power offset of the PCH is specified by the NRDUCellChnPwr.PagingMaxPwrOffset parameter.

Channel or Signal	Configuration
PDSCH with signaling transmitted during RRC connection setup in high-frequency SA networking	A power offset of 3 dB is set by enabling signaling radio bearer (SRB) power aggregation (selecting the SRB_PA_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter).
PDCCH and PDSCH with signaling or broadcast messages, such as paging messages, Remaining Minimum System Information (RMSI), other system information (OSI), and synchronization signal and PBCH block (SSB) messages, transmitted during RRC connection setup in high-frequency SA networking with carrier aggregation (CA) enabled	The power offset is increased to a maximum of 3 dB by enabling Inter-component carrier (inter-CC) power aggregation. Inter-CC power aggregation takes effect when all active cells in a high-frequency sector meet all of the following conditions: • The INTERCELL_PA_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter is selected for the cells. • The NRDUCell.DlBandwidth parameter is set to CELL_BW_100M for the cells. • The NRDUCellTrp.MaxEirp parameter must be set to the same value for the cells. • The number of active common cells (with the NRDUCell.SupplementaryCellInd parameter set to NORMAL_CELL) is less than or equal to the number of active secondary cells (SCells) (with the NRDUCell.SupplementaryCellInd parameter set to ULDL_SUPPLEMENTARY_CELL). You are advised to configure the frequencies of common cells in center bands. The NRDUCell.DlNarfcn parameter must be set to the same value for intra-base-station inter-sector and inter-base-station active common cells in high frequency bands.

 NOTE

- Reconfiguring the preceding parameters used to specify the power offset may interrupt UE services for a short time.
- Unlike LTE, NR does not involve cell-specific reference signal (CRS). For this reason, CRS power cannot be configured.

4.1.2 Dynamic Power Control

4.1.2.1 PDCCH Dynamic Power Control

Currently, PDCCH dynamic power control applies only to cell edge users (CEUs). PDCCH power control for CEUs enables the gNodeB to adaptively adjust the transmit power of PDCCHs dedicated to CEUs based on the PDCCH control

channel element (CCE) aggregation level and PUCCH discontinuous transmission (DTX) feedback.

In low frequency bands, PDCCH power control for CEUs is controlled by the **NRDUCellPdcchAlgo.PdcchEdgePwrCtrlPolicy** parameter.

- If this parameter is set to **NOT_CONFIG**, PDCCH power control for CEUs does not take effect. This value is recommended when the success rate of PDCCH CCE allocation needs to be increased.
- If this parameter is set to **BASED_ON_AGG_LVL**, PDCCH power control for CEUs takes effect. The base station adjusts the PDCCH power based on the PDCCH CCE aggregation level and PDCCH detection result. This value is recommended when the indicators related to accessibility, retainability, and mobility need to be improved.
- If this parameter is set to **BASED_ON_CSI_RPT**, PDCCH power control for CEUs takes effect. Different from when this parameter is set to **BASED_ON_AGG_LVL**, the base station adjusts the PDCCH power based on not only the PDCCH CCE aggregation level and PDCCH detection result but also the CSI reports from UEs, increasing the accuracy of PDCCH power adjustment. This value is recommended when the success rate of PDCCH CCE allocation as well as the indicators related to accessibility, retainability, and mobility need to be considered.

4.1.2.2 PDSCH Dynamic Power Control

PDSCH dynamic power control includes UE power and bandwidth adaptation, power aggregation in unoccupied bandwidths, power reduction for CCUs, and PDSCH DMRS power adaptation.

UE Power and Bandwidth Adaptation

UEs using extremely low MCS indexes experience reduced transmission rates. To increase the downlink transmission rates for such UEs, UE power and bandwidth adaptation is introduced. This function adaptively adjusts the PDSCH power spectral density (PSD) based on the MCS and bandwidth for PDSCH scheduling, thereby increasing transmission rates.

The maximum PSD adjustment value for the PDSCH is determined by the **NRDUCellChnPwr.MaxPdschConvPwrOffset** parameter. This function takes effect only for UEs using extremely low MCS indexes, which refer to MCS 0 and MCS 1 in the MCS index table for 256QAM, or MCS 0, MCS 1, and MCS 2 in the MCS index table for 64QAM. For other UEs, UE power and bandwidth adaptation does not take effect, bringing no gains. This function is supported only in low frequency bands, not in high frequency bands.

NOTE

The transmit PSD refers to the transmit power of an RB.

Power Aggregation in Unoccupied Bandwidths

If there are unoccupied RBs in the current bandwidth during scheduling, the power allocated to this bandwidth is not fully utilized. To address this issue, power aggregation in unoccupied bandwidths is introduced. This function adaptively

adjusts the PDSCH PSD based on the remaining power after scheduling. It aggregates and reallocates the unused power from unoccupied RBs to other RBs, thereby increasing downlink transmission rates. The maximum PSD adjustment value for the PDSCH is determined by the **NRDUCellChnPwr.MaxPdschConvPwrOffset** parameter. This function is supported only in low frequency bands, not in high frequency bands.

Power Reduction for CCUs

Power reduction for CCUs adaptively reduces the PDSCH PSD based on the rank and MCS for PDSCH scheduling or UEs' received signal strength estimated by the base station. (CCU is short for cell center user.) This function takes effect in either of the following scenarios:

- Scenario 1: The base station reduces the PDSCH PSD to optimize the error vector magnitude (EVM) as follows:
 - Usually, this happens when the MCS index for rank 2 exceeds the low-rank MCS index threshold for base station EVM or the MCS index for rank 3 or 4 exceeds the high-rank MCS index threshold for base station EVM.
 - The reduction value is determined by the **NRDUCellChnPwr.MaxBsEvmPwrOffset** parameter.
 - The low-rank MCS index threshold for base station EVM is specified by the **NRDUCellChnPwr.BsEvmLowRankMcsThld** parameter.
 - The high-rank MCS index threshold for base station EVM is specified by the **NRDUCellChnPwr.BsEvmHighRankMcsThld** parameter.
 - Power reduction for CCUs using 1024QAM is controlled by the **NRDUCellChnPwr.Max1024QamPwrOffset** parameter.
 - If this parameter is set to **0**, power reduction for CCUs takes effect as usual.
 - If this parameter is set to a value other than **0** and an MCS enabling 1024QAM is used, the base station reduces the PDSCH PSD by an amount specified by the **NRDUCellChnPwr.Max1024QamPwrOffset** parameter.

For details about 1024QAM, see *Modulation Schemes*.

- Scenario 2: The base station reduces the PDSCH PSD to prevent UE power saturation when the estimated received signal strength of a UE exceeds the sum of the value defined in 3GPP specifications and the UE power saturation offset, and the rank value is greater than or equal to the rank threshold for UE power saturation.
 - The maximum reduction value is determined by the **NRDUCellChnPwr.MaxUeSatPwrOffset** parameter.
 - The rank threshold for UE power saturation is specified by the **NRDUCellChnPwr.UeSatPwrRankThld** parameter.
 - The UE power saturation offset for UEs prone to power saturation is specified by the **NRDUCellChnPwr.SpecUeRssiSatOffset** parameter.
 - The UE power saturation offset for other UEs is specified by the **NRDUCellChnPwr.UeSatPwrOffsetThld** parameter.

For details about the received signal strength value defined in 3GPP specifications, see Table 7.4-1: Maximum input level in 3GPP TS 38.101-1 V15.6.0.

When both scenarios take effect, the reduction in the PDSCH PSD is determined by the larger value between the reduction value in scenario 1 (which equals to **NRDUCellChnPwr.MaxBsEvmPwrOffset** or **NRDUCellChnPwr.Max1024QamPwrOffset**) and the actual reduction value in scenario 2 (which may be less than **NRDUCellChnPwr.MaxUeSatPwrOffset**).

This function is supported only in low frequency bands, not in high frequency bands.

PDSCH DMRS Power Adaptation

PDSCH DMRS power adaptation is introduced to adaptively increase the PDSCH DMRS PSD based on the rank and MCS for scheduling, improving the demodulation performance of CEUs. This function is controlled by the **DMRS_PWR_ADAPT_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter.

This document describes only this function in low frequency bands. For details about this function in high frequency bands, see *mmWave Super Coverage (High-Frequency TDD)*.

4.2 Network Analysis

4.2.1 Benefits

The gNodeB implements static and dynamic power control for downlink channels or signals. Power reduction for CCUs can increase the throughput of CCUs, while static power control, UE power and bandwidth adaptation, power aggregation in unoccupied bandwidths, and PDSCH DMRS power adaptation can improve the coverage performance of corresponding downlink channels or signals for CEUs.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

For static power control, a higher power offset for downlink channels or signals produces more coverage gains, but generates more interference to neighboring cells. This may decrease the data channel power and downlink user rates.

For PDCCH dynamic power control, a higher proportion of CEUs in a cell indicates a higher proportion of UEs for which dynamic power control takes effect and more PDCCH coverage gains, but fewer available PDCCH CCEs and more interference to neighboring cells.

When the number of NR carriers configured for an RF module is changed from two to more than two, the base station will experience a smaller power increase. In this case, the cell coverage may change, resulting in a lower access success rate, higher service drop rate, worse user experience at the coverage edge, and reduced benefits provided by power control.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	The following functions take effect only when rate-matching-pattern-configuration-based PDCCH rate matching is disabled. • UE power and bandwidth adaptation • Power aggregation in unoccupied bandwidths
Low-frequency TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDLAmc.DLModulationSchemeSw parameter	<i>Modulation Schemes</i>	Power reduction for CCUs using 1024QAM takes effect only when DL 1024QAM is enabled.
High-frequency TDD	None	None	None	None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PEI	PEI_SW option of the NRDUCellUePwrSaving.IdeUePwrSavingSwitch parameter	<i>UE Power Saving</i>	When the NRDUCellChnPwr.r.MaxCommonDcI PwrOffset parameter is set to a value greater than 0 , PEI cannot be enabled.

4.3.2.3 Function Impacts

- Functions related to power sharing

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When both power reduction for CCUs in scenario 2 ^a and LTE TDD and NR Flash Dynamic Power Sharing are enabled, the benefits provided by flash dynamic power sharing on the NR side may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When the maximum SSB power offset is set to a value greater than 3 dB and LTE TDD and NR Flash Dynamic Power Sharing is in effect, the SSB power aggregation capability decreases.
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When both power reduction for CCUs in scenario 2 ^a and NR inter-carrier dynamic power sharing are enabled, the benefits provided by NR inter-carrier dynamic power sharing may decrease.
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When both NR inter-carrier dynamic power sharing and power aggregation in unoccupied bandwidths are enabled, the benefits provided by power aggregation in unoccupied bandwidths may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When the maximum SSB power offset is set to a value greater than 3 dB and NR inter-carrier dynamic power sharing is in effect, the SSB power aggregation capability decreases. For the AAU5368md, if the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is selected, NR inter-carrier dynamic power sharing does not work in slots with SSB transmission. This prevents the decrease in the SSB power aggregation capability, but this power sharing function no longer delivers gains in these slots.
Low-frequency TDD	Inter-BWP power sharing	INTER_BWP_POWER_SHARING_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>RedCap</i>	When both PDCCH power control for CEUs and inter-BWP power sharing are enabled, the power consumption of individual UEs increases, leading to power insufficiency in the cell. This, in turn, decreases the CCE allocation success rate and may reduce overall benefits.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap power aggregation	REDCAP_POWER_AGGREGATION_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>RedCap</i>	When both PDCCH power control for CEUs and RedCap power aggregation are enabled, the power consumption of individual UEs increases, leading to power insufficiency in the cell. This, in turn, decreases the CCE allocation success rate and may reduce overall benefits.
a: Power reduction for CCUs is applicable in two scenarios, as described in Power Reduction for CCUs in 4.1.2.2 PDSCH Dynamic Power Control .				

- Functions related to power adjustment

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power aggregation in unoccupied bandwidths	NRDUCellChnPower.MaxPdschCcnvPwrOffset	<i>Power Control</i>	When both power reduction for CCUs in scenario 2 ^a and power aggregation in unoccupied bandwidths are enabled, the benefits provided by power aggregation in unoccupied bandwidths may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	- When LTE and NR multi-band power adaptive scheduling and power reduction for CCUs are both enabled, the benefits of power adaptive scheduling may decrease on the NR side. - When the NRDUCellTrpBeam.MaxSsbPwrOffset parameter is set to a value greater than 3 dB and LTE and NR multi-band power adaptive scheduling is in effect: - If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPower.ChnPwrAlgoSwitch parameter is deselected, the SSB power aggregation capability decreases. - If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPower.ChnPwrAlgoSwitch parameter is selected, the SSB power aggregation capability does

RAT	Function Name	Function Switch	Reference	Description
				not decrease. This is because, with this option selected, LTE and NR multi-band power adaptive scheduling will not be performed (and will not provide gains) in slots with SSB transmission.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When both NR multi-band power adaptive scheduling and power aggregation in unoccupied bandwidths are enabled, the benefits provided by power aggregation in unoccupied bandwidths may decrease.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When both NR multi-band power adaptive scheduling and power reduction for CCUs are enabled, the benefits provided by power adaptive scheduling may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When the NRDUCellTrpBeam.MaxSsbPwrOffset parameter is set to a value greater than 3 dB and NR multi-band power adaptive scheduling is in effect: - If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is deselected, the SSB power aggregation capability decreases. - If the PWR_SHR_SSB_PWR_AGG_TD_SCH_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter is selected, the SSB power aggregation capability does not decrease. This is because, with this option selected, NR multi-band power adaptive scheduling will not be performed (and will not provide gains) in slots with SSB transmission.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink power reduction for low-speed cells in interference coordination between public and dedicated networks	NRDUCellHighSpeedMob.DlInterfBasedPwrOffset	<i>High Speed Mobility</i>	As downlink power reduction for low-speed cells in interference coordination between public and dedicated networks aims to reduce interference to high-speed cells, power aggregation in unoccupied bandwidths does not take effect when downlink power reduction for low-speed cells is enabled.
Low-frequency TDD	L/NR Inter-band Deep Coordination	LTE: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter NR: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShareCfg.DynPwrShareAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When both power reduction for CCUs and L/NR Inter-band Deep Coordination are enabled, the benefits provided by power sharing on the NR side may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L/NR Inter-band Deep Coordination	LTE: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter NR: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShareCfg.DynPwrShareAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When the NRDUCellTrpBeam.MaxSsbPwrOffset parameter is set to a value greater than 3 dB and L/NR Inter-band Deep Coordination is in effect, the SSB power aggregation capability decreases.
a: Power reduction for CCUs is applicable in two scenarios, as described in Power Reduction for CCUs in 4.1.2.2 PDSCH Dynamic Power Control .				

- Others

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	While intra-base-station DL CoMP is in effect, power reduction for CCUs can take effect in cooperating cells only when the SRS RSRP difference between the serving and cooperating cells is less than or equal to 6 dB. This increases the downlink throughput of UEs involved in intra-base-station DL CoMP.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI-RS rate matching	CSIRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When power reduction for CCUs takes effect in low frequency bands, enabling CSI-RS rate matching may cause an unexpectedly high block error rate (BLER) in the slot with CSI-RS transmission. For this reason, it is recommended that the NZP_CSIRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter be deselected in such scenarios. Once deselected, REs that are not used for non-zero-power CSI-RS (NZP CSI-RS) transmission on the symbols to which NZP CSI-RS resources are mapped are not used for data scheduling.
Low-frequency TDD	Fusion Cell	NRDUCell.NrDualCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	It is recommended that the maximum transmit power, maximum SSB power offset, TRS power offset, and maximum value of the aggregated power offset on the PDSCH be consistent between the two combined AAUs. Otherwise, the gains of Fusion Cell may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QAMSwitch	<i>Modulation Schemes</i>	If there are UEs for which DL 256QAM is being used, the proportion of UEs for which power aggregation in unoccupied bandwidths takes effect may decrease.
Low-frequency TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDLAmc.DLModulationSchemeSw parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, the proportion of UEs for which power aggregation in unoccupied bandwidths takes effect may decrease.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	When both PDCCH power control for CEUs and RedCap Introduction are enabled, the power consumption of individual UEs increases, leading to power insufficiency in the cell. This, in turn, decreases the uplink and downlink CCE allocation success rates as well as the average uplink and downlink throughputs of RedCap UEs.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

The following RF modules support downlink power control. For details about these RF modules, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

- All NR TDD-capable 5000 series RRUs and AAUs
- All NR TDD-capable non-5000 series RRUs and AAUs with 32T32R or higher
- All NR TDD-capable mmWave modules
- All NR TDD-capable pRRUs (However, the pRRU565 xAB , pRRU575 xAB , and pRRU585 xAB series modules do not support static power control. x is a digit; A and B stand for different letters.)

Other modules do not support downlink power control and use the average transmit power over the entire bandwidth (cell reference power).

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 4-3](#) and [Table 4-4](#) describe the parameters used for optimization of static and dynamic downlink power control, respectively.

Table 4-3 Parameters used for optimization of static power control

Parameter Name	Parameter ID	Setting Notes
Max Transmit Power	NRDUCellTrp.MaxTransmitPower	This parameter is mandatory for low-frequency cells served by macro base stations and optional for cells served by LampSite base stations. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Max Transmit Power in Milliwatt	NRDUCellTrp.MaxTransmitPowerMw	Set this parameter based on the network plan. This parameter is valid and mandatory only when the NRDUCellTrp.PowerConfigMode parameter is set to TRANSMIT_POWER_MW .
Power Config Mode	NRDUCellTrp.PowerConfigMode	Set this parameter to TRANSMIT_POWER or TRANSMIT_POWER_MW for low-frequency cells and to EIRP for high-frequency cells. This parameter does not take effect when the NRDUCellTrp.TxRxMode parameter is set to 2R or 4R .
Max EIRP	NRDUCellTrp.MaxEirp	This parameter is mandatory for high-frequency cells served by macro base stations. Set this parameter based on the network plan. When inter-CC power aggregation is enabled, the NRDUCellTrp.MaxEirp parameter must be set to the same value for all active cells in a high-frequency sector.
Max Transmit Power	NRDUCellCoverage.MaxTransmitPower	This parameter is optional for cells served by LampSite base stations. Set this parameter based on the network plan.
Max Transmit Power in Milliwatt	NRDUCellCoverage.MaxTransmitPowerMw	Set this parameter based on the network plan. This parameter is valid and mandatory only when the NRDUCellCoverage.PowerConfigMode parameter is set to TRANSMIT_POWER_MW .
Power Config Mode	NRDUCellCoverage.PowerConfigMode	Set this parameter to TRANSMIT_POWER_DBM or TRANSMIT_POWER_MW for low-frequency cells.
Max SSB Power Offset	NRDUCellTrpBeam.MaxSsbPwrOffset	This parameter is mandatory for low-frequency NR TDD cells. It is recommended that Huawei engineers determine the offset value based on the live network.

Parameter Name	Parameter ID	Setting Notes
Max SSB Power Offset	NRDUCellTrpMmwavBeam.MaxSsbPwrOffset	This parameter is mandatory for high-frequency NR TDD cells. It is recommended that Huawei engineers determine the offset value based on the live network.
Maximum Common DCI Power Offset	NRDUCellChnPwr.MaxCommonDciPwrOffset	It is recommended that Huawei engineers determine the offset value based on the live network.
TRS Power Offset	NRDUCellChnPwr.TrsPwrOffset	It is recommended that Huawei engineers determine the offset value based on the live network.
Max Paging Power Offset	NRDUCellChnPwr.PagingMaxPwrOffset	Set this parameter in high frequency bands. The value 0 is recommended.
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	It is recommended that the SRB_PA_SW option be selected in high frequency bands.
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	Set this parameter in high frequency bands. It is recommended that the INTERCELL_PA_SW option be selected by Huawei engineers based on the live network.
Supplementary Cell Indicator	NRDUCell.SupplementaryCellInd	Set this parameter in high frequency bands. It is recommended that Huawei engineers determine the value based on the live network.
Downlink NARFCN	NRDUCell.DlNarfcn	Set this parameter in high frequency bands. The NRDUCell.DlNarfcn parameter must be set to the same value for inter-base-station active common cells (with the NRDUCell.SupplementaryCellInd parameter set to NORMAL_CELL) in high frequency bands. It is recommended that Huawei engineers determine the value based on the live network.

Table 4-4 Parameters used for optimization of dynamic power control

Parameter Name	Parameter ID	Setting Notes
Maximum Converged PDSCH Power	NRDUCellChnPwr.MaxPdschConvPwrOffset	It is recommended that Huawei engineers determine the offset value based on the live network.
Maximum Base Station EVM Power Offset	NRDUCellChnPwr.MaxBsEvmPwrOffset	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the offset value based on the live network.
Maximum 1024QAM Power Offset	NRDUCellChnPwr.Max1024QamPwrOffset	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the offset value based on the live network.
Base Station EVM High Rank MCS Threshold	NRDUCellChnPwr.BsEvmHighRankMcsThld	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the value based on the live network.
Base Station EVM Low Rank MCS Threshold	NRDUCellChnPwr.BsEvmLowRankMcsThld	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the value based on the live network.
Maximum UE Saturation Power Offset	NRDUCellChnPwr.MaxUeSatPwrOffset	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the offset value based on the live network. If there is a large proportion of SSBs with RSRP higher than -65 dBm (saturation value), a large value of this parameter is recommended. If there is a smaller proportion of SSBs with RSRP higher than -65 dBm, a small value of this parameter is recommended.
UE Saturation Power Offset Threshold	NRDUCellChnPwr.UeSatPwrOffsetThld	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the value based on the live network.
Specific UE RSSI Saturation Offset	NRDUCellChnPwr.SpecUeRssiSatOffset	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the value based on the live network.

Parameter Name	Parameter ID	Setting Notes
UE Saturation Power Rank Threshold	NRDUCellChnPwr.UeSatPwrRankThld	Set this parameter in low frequency bands. It is recommended that Huawei engineers determine the value based on the live network.
Channel Power Algorithm Switch	NRDUCellChnPwr.ChnPwrAlgoSwitch	Set this parameter in low frequency bands. It is recommended that the DMRS_PWR_ADAPT_SW option be selected by Huawei engineers based on the live network.
PDCCH Edge Power Control Policy	NRDUCellPdcchAlgo.PdcchEdgePwrCtrlPolicy	Set this parameter to NOT_CONFIG in extremely heavy-load scenarios (for example, when more than 600 UEs access a cell per second). Set this parameter to BASED_ON_CSI_RPT when PDCCH resources are insufficient (downlink or uplink CCE allocation failure rate is greater than 30%). Set this parameter to BASED_ON_AGG_LVL in other scenarios.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

For low-frequency TDD cells served by macro base stations

```
//Setting the maximum transmit power of the NR DU cell TRP to 30.9 dBm
MOD NRDUCELLTRP: NrDuCellTrpId=0, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=309;
//Configuring the maximum SSB power offset
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=0, CoverageScenario=DEFAULT, MaxSsbPwrOffset=0;
//Configuring the power offset of DCI in common search space
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxCommonDciPwrOffset=0;
```

```
//Configuring the TRS power offset
MOD NRDUCELLCHNPWR: NrDuCellId=0, TrsPwrOffset=0;
//Configuring the maximum converged PDSCH power offset
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxPdschConvPwrOffset=0;
//Configuring the MaxBsEvmPwrOffset parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxBsEvmPwrOffset=3;
//Configuring the Max1024QamPwrOffset parameter for power reduction for CCUs using 1024QAM
MOD NRDUCELLCHNPWR: NrDuCellId=0, Max1024QamPwrOffset=3;
//Configuring the BsEvmHighRankMcsThld parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, BsEvmHighRankMcsThld=21;
//Configuring the BsEvmLowRankMcsThld parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, BsEvmLowRankMcsThld=25;
//Configuring the MaxUeSatPwrOffset parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxUeSatPwrOffset=3;
//Configuring the UeSatPwrOffsetThld parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, UeSatPwrOffsetThld=78;
//Configuring the SpecUeRssiSatOffset parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, SpecUeRssiSatOffset=78;
//Configuring the UeSatPwrRankThld parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, UeSatPwrRankThld=RANK1;
//Enabling PDSCH DMRS power adaptation
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=DMRS_PWR_ADAPT_SW-1;
//Enabling PDCCH power control for CEUs
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, PdcchEdgePwrCtrlPolicy=BASED_ON_AGG_LVL;
```

For high-frequency cells served by macro base stations

```
//Setting the maximum transmit power of the NR DU cell TRP to 30.9 dBm
MOD NRDUCELLTRP: NrDuCellTrpId=0, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=309;
//Setting the maximum EIRP of the NR DU cell TRP to 59.0 dBm
MOD NRDUCELLTRP: NrDuCellTrpId=0, PowerConfigMode=EIRP, MaxEirp=590;
//Configuring the maximum SSB power offset
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=0, CoverageScenario=DEFAULT, MaxSsbPwrOffset=0;
//Configuring the power offset of DCI in common search space
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxCommonDciPwrOffset=0;
//Configuring the TRS power offset
MOD NRDUCELLCHNPWR: NrDuCellId=0, TrsPwrOffset=0;
//Configuring the maximum converged PDSCH power offset
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxPdschConvPwrOffset=0;
//Configuring the PagingMaxPwrOffset parameter
MOD NRDUCELLCHNPWR: NrDuCellId=0, PagingMaxPwrOffset=0;
//Enabling SRB power aggregation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, FullChannelCovEnhSwitch=SRB_PA_SW-1;
//Enabling inter-CC power aggregation when at least two cells are activated
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, FullChannelCovEnhSwitch=INTERCELL_PA_SW-1;
//Configuring half of the activated cells as uplink and downlink supplementary cells when at least two cells
are activated (If an odd number of cells are activated, round up half the number.)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, FrequencyBand=N257,
SupplementaryCellInd=ULDL_SUPPLEMENTARY_CELL;
```

For cells served by LampSite base stations

```
//Setting the maximum transmit power of the NR DU cell TRP to 30.9 dBm
MOD NRDUCELLTRP: NrDuCellTrpId=0, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=309;
//Setting the maximum transmit power in the coverage of the NR DU cell to 30.9 dBm
MOD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0,
PowerConfigMode=TRANSMIT_POWER_DBM, MaxTransmitPower=309;
//Configuring the maximum SSB power offset
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=0, CoverageScenario=DEFAULT, MaxSsbPwrOffset=0;
//Configuring the power offset of DCI in common search space
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxCommonDciPwrOffset=0;
//Configuring the TRS power offset
MOD NRDUCELLCHNPWR: NrDuCellId=0, TrsPwrOffset=0;
//Configuring the maximum converged PDSCH power offset
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxPdschConvPwrOffset=0;
//Configuring the MaxBsEvmPwrOffset parameter for power reduction for CCUs
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxBsEvmPwrOffset=3;
//Configuring the Max1024QamPwrOffset parameter for power reduction for CCUs using 1024QAM
MOD NRDUCELLCHNPWR: NrDuCellId=0, Max1024QamPwrOffset=3;
//Configuring the BsEvmHighRankMcsThld parameter for power reduction for CCUs
```

```
MOD NRDUCELLCHNPWR: NrDuCellId=0, BsEvmHighRankMcsThld=21;  
//Configuring the BsEvmLowRankMcsThld parameter for power reduction for CCUs  
MOD NRDUCELLCHNPWR: NrDuCellId=0, BsEvmLowRankMcsThld=25;  
//Configuring the MaxUeSatPwrOffset parameter for power reduction for CCUs  
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxUeSatPwrOffset=3;  
//Configuring the UeSatPwrOffsetThld parameter for power reduction for CCUs  
MOD NRDUCELLCHNPWR: NrDuCellId=0, UeSatPwrOffsetThld=78;  
//Configuring the SpecUeRssiSatOffset parameter for power reduction for CCUs  
MOD NRDUCELLCHNPWR: NrDuCellId=0, SpecUeRssiSatOffset=78;  
//Configuring the UeSatPwrRankThld parameter for power reduction for CCUs  
MOD NRDUCELLCHNPWR: NrDuCellId=0, UeSatPwrRankThld=RANK1;  
//Enabling PD SCH DMRS power adaptation  
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=DMRS_PWR_ADAPT_SW-1;  
//Enabling PDCCH power control for CEUs  
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, PdcchEdgePwrCtrlPolicy=BASED_ON_AGG_LVL;
```

Deactivation Command Examples

MML commands are not required for deactivating this function.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

None

4.4.3 Network Monitoring

None

5 Uplink Power Control

Uplink power control enables the gNodeB to calculate the TPC value based on the power control parameter settings, neighboring cell interference, measurement results of uplink data transmission, and feedback from the UE and then issue a TPC command to the UE through the PDCCH. In accordance with the mapping defined in chapter 7 "Uplink Power control" in 3GPP TS 38.213 V15.6.0, the UE converts the TPC command to the power adjustment amount. Then, the UE determines the uplink transmit power based on factors including its own maximum transmit power, the nominal power of the cell, the path loss, the MCS, and the amount of radio resources.

The maximum transmit power of a UE differs in SA networking and NSA networking as follows:

- In SA networking, as the UE transmits signals only on the NR side, its maximum transmit power (P_{CMAX}) is determined by its capability.
- In NSA networking:
 - In low frequency bands, the RF channels for the UE on the NR and LTE sides are not independent. Power sharing is implemented when signals are transmitted on both sides. The maximum uplink transmit power P_{NR} and P_{LTE} are configured for the UE on the NR and LTE sides, respectively. The sum of P_{NR} and P_{LTE} cannot exceed P_{CMAX} . In this case, the maximum transmit power of the UE on the NR and LTE sides cannot exceed P_{NR} and P_{LTE} , respectively.
 - In high frequency bands, the RF channels for the UE on the NR and LTE sides are independent, which do not require power sharing. Therefore, the maximum transmit power of the UE on both the NR and LTE sides is P_{CMAX} .

Uplink power control is applicable to all uplink channels and signals, including:

- Physical random access channel (PRACH)
- Physical uplink shared channel (PUSCH)
- Physical uplink control channel (PUCCH)
- Sounding reference signal (SRS)

The gNodeB delivers the power adjustment amount (through a TPC command) during power control for all uplink channels and signals, except for PRACH.

 NOTE

Unless otherwise specified, in this chapter, the signal to interference plus noise ratio (SINR) and reference signal received power (RSRP) refer to the SINR and RSRP of a specific uplink channel or signal.

5.1 PRACH Power Control

5.1.1 Principles

If the PRACH power is too low, the random access success rate decreases. If the PRACH power is too high, the interference to neighboring cells and the UE power consumption increase. To address this, PRACH power control is introduced to minimize the preamble transmit power while ensuring the random access success rate. As a result, interference to neighboring cells reduces and power consumption of UEs decreases.

During PRACH power control, the gNodeB does not deliver TPC commands, and the corresponding PRACH power is determined by [PRACH Power Calculation](#). The PRACH power control process consists of initial power determination and power ramping.

- Initial power determination (see [PRACH Power Calculation](#)): The UE determines the uplink transmit power based on such information as the path loss and the power control parameters initially delivered by the gNodeB. The power is the minimum of either the maximum transmit power of the UE or the calculated power.
- Power ramping (see [Power Ramping](#)): If the UE fails in random access and certain conditions are met, the PRACH transmit power can be gradually increased to ensure that the minimum power is used for a successful random access.

PRACH Power Calculation

The PRACH transmit power is calculated using the following formula:

$$P_{\text{PRACH}} = \min(P_{\text{CMAX}}, P_{\text{o_pre}} + \text{PL} + \Delta_{\text{preamble}} + (N_{\text{pre}} - 1) \cdot \Delta_{\text{step}})$$

where

- P_{CMAX} is the maximum transmit power of the UE.
- $P_{\text{o_pre}}$ is the target power expected by the gNodeB when the requirements for the preamble detection performance are met and the PRACH preamble format is 0. This variable is specified by the **NRDUCellUPLPcConfig.PreambleInitRxTargetPwr** parameter.
- PL is the downlink path loss estimated by the UE. This value is obtained based on the measured RSRP value of the downlink channel and the SSB transmit power.
- Δ_{preamble} is the offset of the power for the current preamble format relative to the power for preamble format 0.

- N_{pre} is the number of times the UE sends preambles. This value cannot exceed the maximum number of preamble transmissions that is specified by the **NRDUCellPrach.MaxPreambleTransCnt** parameter. When determining the initial PRACH power, the value of N_{pre} is 1.
- Δ_{step} is the preamble power ramping step. This variable is specified by the **NRDUCellULPcConfig.PwrRampingStep** parameter.

P_{o_pre} and Δ_{step} are delivered by the gNodeB to a UE through SIB1 (in SA networking) or RRC signaling (in NSA networking). The UE calculates the transmit power for the random access preamble (PRACH transmit power) according to the preceding formula.

Power Ramping

If a random access procedure initiated by a UE fails before the number of preamble transmissions reaches the upper limit, the UE retransmits the random access preamble using increased PRACH transmit power. The overall PRACH transmit power ramping process is as follows:

1. The UE sends a random access preamble.
2. If the gNodeB correctly detects this preamble, it sends a random access response (RAR) to the UE.
3. If the UE does not receive the RAR, it retransmits the preamble with the PRACH transmit power increased by Δ_{step} each time, and counts N_{pre} until either an RAR is received or N_{pre} reaches the maximum number of preamble transmissions.

5.1.2 Network Analysis

5.1.2.1 Benefits

PRACH power control minimizes the preamble transmit power while ensuring the random access success rate. As a result, interference to neighboring cells reduces and power consumption of UEs decreases.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

PRACH power control determines the power of a UE in the access phase, directly affecting the access success rate, followed by the handover success rate and service drop rate.

5.1.3 Requirements

5.1.3.1 Licenses

None

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

None

5.1.3.2.2 Mutually Exclusive Functions

None

5.1.3.2.3 Function Impacts

None

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.3.4 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 5-1](#) describes the parameters used for optimization.

Table 5-1 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Power Ramping Step	NRDUCellULPCConfig.PwrRampingStep	Set this parameter based on the network plan.
Maximum Preamble Transmission Count	NRDUCellPrach.MaxPreambleTransCnt	Set this parameter based on the network plan.
Preamble Initial Received Target Power	NRDUCellULPCConfig.PreambleInitRxTargetPwr	Set this parameter based on the network plan.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Configuring the preamble power ramping step
MOD NRDUCELLULPCONFIG: NrDuCellId=0, PwrRampingStep=DB4;
//Configuring the maximum number of preamble transmissions
MOD NRDUCELLPRACH: NrDuCellId=0, MaxPreambleTransCnt=N10;
//Configuring the initial receive target power of the preamble for the cell
MOD NRDUCELLULPCONFIG: NrDuCellId=0, PreambleInitRxTargetPwr=-50;
```

Deactivation Command Examples

MML commands are not required for deactivating this function.

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

None

5.1.4.3 Network Monitoring

None

5.2 PUSCH Power Control

5.2.1 Principles

PUSCH power control is introduced to reduce the interference to neighboring cells and improve the cell traffic volume, while ensuring the data rates of CEUs. It allows UEs to adjust their transmit power on the PUSCH based on the TPC commands received from the gNodeB. For details, see [5.2.1.1 Basic Power Control for PUSCH](#) and [5.2.1.2 Power Control Optimization for PUSCH](#).

5.2.1.1 Basic Power Control for PUSCH

Basic power control for PUSCH includes Msg3 power control during random access, PUSCH power calculation, and power headroom calculation.

- The transmit power for Msg3, the first PUSCH message, is calculated using the formula described in [Msg3 Power Control During Random Access](#).
- The transmit power for the PUSCH is calculated using the formula described in [PUSCH Power Calculation](#).
- Power headroom, available when the PUSCH transmit power does not reach the maximum UE transmit power, is calculated as described in [Power Headroom Calculation](#).

Msg3 Power Control During Random Access

Msg3 in the random access procedure is the RRC Connection Request message. When the PUSCH carries Msg3, the Msg3 transmit power is calculated based on P_{o_pre} and parameters related to PUSCH power control using the following formula:

$$P_{PUSCH}(i) = \min\{P_{CMAX}, 10\log_{10}(2^{\mu} \times M_{PUSCH}(i)) + P_{o_pre} + \Delta_{PREAMBLE_Msg3} + PL + \Delta_{TF}(i) + f(i)\}$$

Table 5-2 defines the variables involved.

Table 5-2 Definitions of the variables in the formula for calculating the Msg3 transmit power

Variable	Definition
i	Indicates the uplink subframe number.
P_{CMAX}	Indicates the maximum transmit power of the UE.
μ	Indicates the subcarrier spacing configuration factor. The subcarrier spacing is $2^{\mu} \cdot 15\text{kHz}$.
$M_{\text{PUSCH}}(i)$	Indicates the number of RBs used in uplink subframe i .
$P_{\text{o_pre}}$	Indicates the target power expected by the gNodeB when the requirements for the preamble detection performance are met and the PRACH preamble format is 0.
$\Delta_{\text{PREAMBLE_Msg3}}$	Indicates the offset of the Msg3 power relative to the random access preamble power during random access. This offset is specified by the NRDUCellUPLCConfig.DeltaPreambleMsg3 parameter.
PL	Indicates the downlink path loss estimated by the UE. This value is obtained based on the measured RSRP value and SSB transmit power.
$\Delta_{\text{TF}}(i)$	Indicates the offset of the power for the MCS in use relative to the power for the reference MCS.
$f(i)$	Indicates the adjustment amount for the PUSCH transmit power of the UE. This value is calculated based on $f(i-1)$ and the TPC information sent over the PDCCH. The initial $f(i)$ is calculated using the following formula: $f(0) = \Delta P_{\text{rampup}} + \sigma_{\text{msg2}}$ where ΔP_{rampup} is the transmit power increase for the random access preamble. σ_{msg2} is the offset used for power calculation. This variable is specified by the NRDUCellUPLCConfig.TpcValueInMsg2 parameter. - For Msg3 power control during random access, $f(i)$ is calculated according to section 7.1 "Physical uplink shared channel" in 3GPP TS 38.213 V18.5.0. Since no DCI containing TPC commands is received during the random access procedure, the TPC value from the PDCCH is not included in the calculation.

PUSCH Power Calculation

The PUSCH transmit power is calculated using the following formula:

$$P_{\text{PUSCH}}(i) = \min\{P_{\text{CMAX},c}, 10 \times \log_{10}(2^{\mu} \times M_{\text{PUSCH}}(i)) + P_{o_PUSCH} + \alpha_{\text{PUSCH}} \times PL + \Delta_{\text{TF}}(i) + f(i)\}$$

After $P_{o_NOMINAL_PUSCH}$, $P_{o_UE_PUSCH}$, σ_{msg2} , and α_{PUSCH} are delivered by the gNodeB to the UE through RRC signaling, the UE calculates the PUSCH transmit power according to the preceding formula. [Table 5-3](#) defines the variables involved.

Table 5-3 Definitions of the variables in the formula for calculating the PUSCH transmit power

Variable	Definition
i	Indicates the uplink slot number.
$P_{\text{CMAX},c}$	Indicates the maximum transmit power of the UE in cell c .
μ	Indicates the subcarrier spacing configuration factor. The subcarrier spacing is $2^{\mu} \cdot 15\text{kHz}$.
$M_{\text{PUSCH}}(i)$	Indicates the number of RBs used by the PUSCH in uplink slot i .
P_{o_PUSCH}	Indicates the received PSD expected by the gNodeB. It is calculated using the following formula: $P_{o_PUSCH} = P_{o_NOMINAL_PUSCH} + P_{o_UE_PUSCH}$ where $P_{o_NOMINAL_PUSCH}$ is the transmit PSD for the PUSCH expected by the gNodeB when correct PUSCH demodulation is ensured. This variable is specified by the NRDUCellUPLCConfig.PoNominalPusch parameter. $P_{o_UE_PUSCH}$ is the power offset of the UE relative to $P_{o_NOMINAL_PUSCH}$. This value reflects the impact of the UE class, service type, and channel quality on the PUSCH transmit power.
α_{PUSCH}	Indicates the path loss compensation factor. This variable is specified by the NRDUCellUPLCConfig.PathLossCoeff parameter.
PL	Indicates the downlink path loss estimated by the UE. This value is obtained based on the measured RSRP value and SSB transmit power.
$\Delta_{\text{TF}}(i)$	Indicates the offset of the power for the MCS in use relative to the power for the reference MCS.

Variable	Definition
$f(i)$	Indicates the adjustment amount for the PUSCH transmit power of the UE. This value is calculated based on $f(i-1)$ and the TPC information sent over the PDCCH. The initial $f(i)$ is calculated using the following formula: $f(0) = \Delta P_{\text{rampup}} + \sigma_{\text{msg2}}$ where ΔP_{rampup} is the transmit power increase for the random access preamble. σ_{msg2} is the offset used for power calculation. This variable is specified by the NRDUCellUPLCConfig.TpcValueInMsg2 parameter.

 NOTE

- If the values of $P_{o_NOMINAL_PUSCH}$, σ_{msg2} , and α_{PUSCH} are too small, the access success rate and service rate of the UE will decrease.
- If the values of $P_{o_NOMINAL_PUSCH}$, σ_{msg2} , and α_{PUSCH} are too large, UEs will be allocated fewer resources and the transmit power during the initial access phase will increase. This results in higher interference to neighbouring cells.

Power Headroom Calculation

Power headroom is available when the PUSCH transmit power does not reach the maximum UE transmit power. It is calculated using the following formula, whose variables are described in [Table 5-3](#).

$$PH_{\text{PUSCH}}(i) = P_{\text{CMAX},c} - (10 \times \log_{10}(2^{\mu} \times M_{\text{PUSCH}}(i)) + P_{o_PUSCH} + \alpha_{\text{PUSCH}} \times PL + \Delta_{\text{TF}}(i) + f(i))$$

After $P_{o_NOMINAL_PUSCH}$, $P_{o_UE_PUSCH}$, and α_{PUSCH} are delivered by the gNodeB to the UE through RRC signaling, the UE calculates the power headroom for PUSCH power control according to the preceding formula.

The UE periodically reports the power headroom to the gNodeB through power headroom reports (PHRs). After receiving a PHR, the gNodeB updates the TPC command delivered to the UE for next transmission, thereby adjusting the PUSCH power. The PHR reporting period is determined by the **phr-PeriodicTimer** specified by the **NRDUCellUPLCConfig** parameter, and the **phr-ProhibitTimer** specified by the **NRDUCellUPLCConfig** parameter.

5.2.1.2 Power Control Optimization for PUSCH

Power control optimization for PUSCH is applied in the following scenarios:

- To adapt to changes in the cell environment, the PUSCH power can be adjusted dynamically (see [PUSCH Transmit Power Adjustment](#)).

- To control interference to neighboring cells, PUSCH joint power control is introduced (see [PUSCH Joint Power Control](#)).
- To increase the cell access success rate in some scenarios, the following functions are introduced:
 - [Simplified Msg4 Power Control Configuration During Random Access](#)
 - [Signaling Power Control Optimization During Random Access](#)
 - [PUSCH Power Control Optimization for Random Access](#)

PUSCH Transmit Power Adjustment

When the **PUSCH_DYN_SCH_CLOSED_LOOP_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch** parameter is selected, the gNodeB adopts closed-loop power control to adaptively adjust the PUSCH transmit power based on the estimated transmit PSD of UEs as well as the RB-level RSRP and SINR, thereby adapting to changes in the PUSCH channel environment and service load during service provisioning.

The PUSCH transmit power adjustment amount corresponds to $f(i)$ in the PUSCH transmit power calculation formula. When the **PUSCH_DYN_SCH_CLOSED_LOOP_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch** parameter is deselected, the gNodeB does not adjust the PUSCH transmit power or deliver TPC commands to UEs. In this case, the value of $f(i)$ remains unchanged.

The preset RB-level RSRP threshold is specified by the **NRDUCellULPcConfig.PuschRsrpThld** parameter. The transmit power of a UE can be increased only when the uplink RSRP of the UE is less than the preset RB-level RSRP threshold.

PUSCH transmit power adjustment also involves the functions described in [Table 5-4](#).

Table 5-4 Functions related to PUSCH transmit power adjustment

RAT	Function Name	Background	Function Switch	Function Description
Low-frequency TDD	PUSCH power reduction for UEs at a medium distance from the cell center	The maximum number of schedulable RBs restricted by power control approaches a specified target value.	PUSCH_MIDP_OINT_RP_SW option of the NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch parameter	When this function is enabled and the maximum number of schedulable RBs restricted by power control is less than the target value, the transmit power of UEs at a medium distance from the cell center is reduced.

RAT	Function Name	Background	Function Switch	Function Description
Low-frequency TDD	Delayed PUSCH power reduction for UEs at a medium distance from the cell center	When a long SRS period is used, changes in the power headroom reported by a UE will cause frequent adjustment of the PUSCH transmit power. As a result, measurement exceptions may occur.	PUSCH_MIDPOINT_RP_DELAY_SW option of the NRDUCellUIMc.UlAmcPerformanceSw parameter	When this function is enabled, the power reduction interval will be prolonged for UEs at a medium distance from the cell center and configured with long SRS periods. This prevents measurement exceptions and improves the accuracy of MCS index selection. This function takes effect only when PUSCH power reduction for UEs at a medium distance from the cell center is enabled.
Low-frequency TDD	Power control optimization for uplink CA	According to the power allocation priority defined in section 7.5 "Prioritizations for transmission power reductions" in 3GPP TS 38.213 V15.9.0, when the power of a UE in the uplink CA state is limited, the primary component carrier (PCC) may occupy too much power and leave no power for secondary component carriers (SCCs), thereby increasing the BLER of SCCs in CA scenarios.	UL_CA_PC_OPT_SW option of the NRDUCellUlpConfig.UlPwrCtrlAlgoSwitch parameter	When this function is enabled, the gNodeB optimizes power allocation for the PCC and SCCs based on the PSD coefficient for uplink CA (specified by the NRDUCellUlpConfig.UlCaPsdCoeff parameter) to relieve power preemption by the PCC and reduce the BLER of SCCs.

RAT	Function Name	Background	Function Switch	Function Description
Low-frequency TDD	Multi-frequency actual-PUSCH-based inner-loop power control	In uplink multi-carrier scenarios, a UE reports two types of PHRs: one for actual PUSCH transmission and another for reference PUSCH transmission. If these two PHRs show significant differences, the TPC value derived from them is inaccurate. This results in an incorrect PUSCH power adjustment, affecting data distribution to SCCs. For details about reference PUSCH transmission, refer to section 7.7.1 "Type 1 PH report" in 3GPP TS 38.213 V15.9.0.	MF_ACTUAL_PUSCH_ILPC_SW option of the NRDUCellUPLCConfig.UPLPwrCtrlAlgoExtSwitch parameter	When this function is enabled, the PUSCH power adjustment amount is calculated solely based on the PHR for actual PUSCH transmission in uplink multi-carrier scenarios. This minimizes changes in the uplink power of non-cell-center UEs and reduces the uplink bit error rate (BER).
Low-frequency TDD	Inter-CC PUSCH power allocation adjustment for uplink CA UEs	In uplink CA scenarios, power may be unevenly allocated among the PCC and SCCs due to factors like abnormal outer loop link adaptation (OLLA) during inter-CC power allocation, affecting the data rates of the PCC and SCCs.	ULCA_INTER_CC_PWR_ALL_OC_ADJ_SW option of the NRDUCellUPLCConfig.UPLPwrCtrlAlgoExtSwitch parameter	When this function is enabled, power allocation is adjusted for the PCC and SCCs, increasing the average uplink UE throughput (User Uplink Average Throughput (DU)).

RAT	Function Name	Background	Function Switch	Function Description
High-frequency TDD	Inter-CC PUSCH power control optimization	In uplink CA scenarios, power control aims to ensure that the total transmit power of a UE does not exceed its capability, and the TPC-based power adjustment amount must be an integer. As a result, the actual total transmit power of the UE may be slightly lower than its maximum transmit power. To ensure that UEs can use their maximum transmit power, this function is introduced.	PUSCH_INTER_CC_PC_OPTION_SW option of the NRDUCellUPLCConfig.UPLPowerCtrlAlgoSwitch parameter	When this function is enabled, power compensation is performed during inter-CC power control. This reduces UE power waste, improves the PUSCH channel quality, and increases the average uplink cell throughput (Cell Uplink Average Throughput (DU)) and average uplink UE throughput (User Uplink Average Throughput (DU)).
High-frequency TDD	Hybrid power control for uplink CA	In uplink CA scenarios, when the path loss decreases, the UE transmit power decreases, and the base station adjusts power with delay. To increase the UE power utilization in such scenarios, this function is introduced.	UL_CA_HYBRID_PC_SW option of the NRDUCellUPLCConfig.UPLPowerCtrlAlgoSwitch parameter	When this function is enabled, the base station uses different PSDs for the PCell and SCells to increase the actual transmit power of UEs, PUSCH channel quality, average uplink cell throughput (Cell Uplink Average Throughput (DU)), and average uplink UE throughput (User Uplink Average Throughput (DU)).

PUSCH Joint Power Control

PUSCH joint power control enables the gNodeB to adjust the PUSCH transmit power of UEs based on the performance of UEs in the serving cell and the

interference to neighboring cells. When the interference to a neighboring cell exceeds a certain level, the gNodeB adjusts the PUSCH transmit power of UEs in the serving cell through TPC commands to ensure the interference to the neighboring cell remains within a certain range. This function is supported only in low frequency bands, not in high frequency bands. It is recommended that PUSCH joint power control be enabled in heavy-load cells or unbalanced-load scenarios where both light-load and heavy-load cells exist.

PUSCH joint power control is controlled by the **PUSCH_JOINT_PC_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch** parameter. To enable this function, both this option and the **PUSCH_MIDPOINT_RP_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch** parameter must be selected.

After PUSCH joint power control is enabled, the serving cell and its neighboring cells exchange their interference information every five minutes. When the interference over thermal (IoT) of a neighboring cell exceeds its IoT threshold specified by the **NRDUCellULPcConfig.PuschJointPcCellIotThld** parameter (for example, the interference from the serving cell is severe), the gNodeB adjusts the PUSCH transmit power of UEs in the serving cell to reduce interference.

To adjust the PUSCH transmit power of UEs, the gNodeB selects target UEs whose transmit power requires adjustment and then decreases the PUSCH transmit power of the selected UEs. The target UEs must meet the following conditions:

- The IoT of PUSCH transmit power is greater than the **NRDUCellULPcConfig.PuschJointPcUserIotThld** parameter value.
- The MCS index is less than the **NRDUCellULPcConfig.PuschJointPcMcsThld** parameter value.
- The UE causes interference to neighboring cells. The gNodeB delivers the measurement configurations for event A3 to the UE, which sends A3 measurement reports when the condition for event A3 is met. Then, the gNodeB determines that the UE causes interference to neighboring cells based on the A3 measurement reports sent by the UE.

Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset for a period of time. A UE reports event A3 if the signal quality meets the condition ($Mn + Ofn + Ocn - Hys > Ms + OfS + Ocs + Off$) throughout *TimeToTrigger*. The variables are described in [Table 5-5](#). For details about event A3, see section 5.5 "Measurements" in 3GPP TS 38.331 V15.5.0.

Table 5-5 Definitions of the variables related to event A3 reporting

Variable	Definition
TimeToTrigger	Indicates a period throughout which the entering condition must be met before the event is reported. The <i>TimeToTrigger</i> for event A3 in PUSCH joint power control is specified by the NRCCellComp.TimeToTrigger parameter.
Ms	Indicates the measurement result of the serving cell.
Mn	Indicates the measurement result of a neighboring cell.

Variable	Definition
Ofs	Indicates the frequency-specific offset of the serving cell. The frequency offsets between intra-frequency cells are 0.
Ofn	Indicates the frequency-specific offset of the neighboring cell. The frequency offsets between intra-frequency cells are 0.
Ocs	Indicates the cell individual offset (CIO) of the serving cell.
Ocn	Indicates the CIO of the neighboring cell. When <i>Ocs</i> is set to 0, <i>Ocn</i> is specified by the NRCellRelation.CellIndividualOffset parameter, which is configured during neighboring cell planning. When this function is enabled, neighboring cell replanning is not required, and therefore, the parameter value does not need to be adjusted.
Hys	Indicates the hysteresis for measurement results. The <i>Hys</i> for event A3 in PUSCH joint power control is specified by the NRCellComp.Hysteresis parameter.
Off	Indicates the offset to measurement results. The <i>Off</i> for event A3 in PUSCH joint power control is specified by the NRCellComp.UlCompRsrpOffset parameter.

Simplified Msg4 Power Control Configuration During Random Access

Simplified Msg4 power control configuration during random access is introduced to increase the access success rate in a cell. After this function is enabled, the Msg5 transmit power on the base station side increases, resulting in a higher Msg5 demodulation success rate. This function is controlled by the **MSG4_PWR_CTRL_CFG_SIMPLIFY_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch** parameter.

Signaling Power Control Optimization During Random Access

Signaling power control optimization during random access is introduced to increase the access success rate in a cell. This function is controlled by the **NRDUCellUIPcConfig.CmpMsgPwrRampStep** parameter.

- When the **NRDUCellUIPcConfig.CmpMsgPwrRampStep** parameter is set to 0, this function does not take effect.
- When the **NRDUCellUIPcConfig.CmpMsgPwrRampStep** parameter is set to a value other than 0, this function takes effect and the power for Msg5 and RRC reconfiguration complete messages in the access phase is increased as follows:
 - If the **MSG5_PWR_CTRL_MCS_SEL_OPT_SW** option of the **NRDUCellUISch.UlSchAlgoSwitch** parameter is deselected, this function takes effect only for RRC reconfiguration complete messages in the access phase.

- If the **MSG5_PWR_CTRL_MCS_SEL_OPT_SW** option of the **NRDUCellUISch.UISchAlgoSwitch** parameter is selected, this function takes effect for both Msg5 and RRC reconfiguration complete messages in the access phase.

PUSCH Power Control Optimization for Random Access

In low-frequency NR TDD, PUSCH power control optimization for random access is introduced to increase the access success rate of CEUs. This function is controlled by the **PUSCH_RA_PC_OPT_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch** parameter and can be enabled only when closed-loop power control for the PUSCH in dynamic scheduling mode (controlled by the **PUSCH_DYN_SCH_CLOSED_LOOP_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch** parameter) is enabled. If the **PUSCH_RA_PC_OPT_SW** option is selected, PUSCH power control optimization for random access is enabled. The base station instructs CEUs to increase the PUSCH transmit power through TPC commands in the access phase. If this option is deselected, PUSCH power control optimization for random access does not take effect.

5.2.2 Network Analysis

5.2.2.1 Benefits

PUSCH power control properly adjusts the PUSCH transmit power to avoid wasting UE power and reduce mutual interference between neighboring cells, while ensuring that the service quality meets requirements in terms of the BLER.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

[Table 5-6](#) describes the network impacts of functions related to PUSCH power control.

Table 5-6 Network impacts of functions related to PUSCH power control

Function	Network Impacts
Basic power control for PUSCH	- PUSCH power control determines the transmit power of a UE and the number of RBs allocated for uplink transmission, thereby affecting uplink throughput. - PUSCH power control affects the transmission of measurement reports and the random access procedure, thereby affecting the handover success rate. - PUSCH power control also affects transmission reliability over uplink channels, affecting the service drop rate. - When downlink data is dynamically distributed at the PDCP layer (with the gNBpdcParamGroup.DlDataPdcpsplit-Mode parameter set to SCG_AND_MCG) in low frequency bands, PUSCH power control causes an unacceptable delay (exceeding 31 ms) for large-packet services. To address this problem, you can enable power control enhancement for uplink preallocation by setting the NRDUCellAlgoSwitch.UlPreallocPwrCtrlEnhSw parameter to ON. However, when UEs in the overlapping area of sectors use MCS 27 (with 256QAM) or MCS 28 (with 64QAM), enabling power control enhancement for uplink preallocation increases interference to neighboring cells. In this scenario, it is recommended that power control enhancement for uplink preallocation be disabled by setting the NRDUCellAlgoSwitch.UlPreallocPwrCtrlEnhSw parameter to OFF.
PUSCH joint power control	PUSCH joint power control has network impacts in the following scenarios: - In unbalanced-load scenarios, the MCS indexes for UEs in cells with power reduction decrease. If no RB resources are available, the throughput of cells with power reduction decreases. - In single-cell scenarios, the BLER of burst services increases and the uplink cell throughput decreases.
Simplified Msg4 power control configuration during random access	After simplified Msg4 power control configuration during random access is enabled, the uplink interference slightly increases.
Signaling power control optimization during random access	After signaling power control optimization during random access is enabled, the larger the NRDUCellUIPcConfig.CmpMsgPwrRampStep parameter value, the stronger the cell interference. The smaller the NRDUCellUIPcConfig.CmpMsgPwrRampStep parameter value, the lower the cell interference.

Function	Network Impacts
Hybrid power control for uplink CA	In high-frequency NR TDD with uplink CA enabled, hybrid power control for uplink CA has the following impacts on the network: • The values of the following indicators may increase: – Average uplink cell throughput (Cell Uplink Average Throughput (DU)) – Average uplink UE throughput (User Uplink Average Throughput (DU)) – Average uplink MCS index – Average uplink rank – Proportion of uplink scheduling occasions using 64QAM – Average uplink interference (N.UL.NI.Avg) – Uplink PRB usage (Uplink Resource Block Utilizing Rate (DU)) • The values of the following indicators may decrease: – Uplink IBLER – Proportion of uplink scheduling occasions using QPSK – Average number of active UEs in a cell (N.User.RRCCConn.Active.Avg) • The values of the following indicators fluctuate depending on UEs' traffic models and UE distribution in the cell: – Proportion of uplink scheduling occasions using 16QAM – Average downlink cell throughput (Cell Downlink Average Throughput (DU)) – Average downlink UE throughput (User Downlink Average Throughput (DU)) – Uplink UE throughput distribution at the RLC layer – Downlink UE throughput distribution at the RLC layer – Average downlink MCS index – Average downlink rank

5.2.3 Requirements

5.2.3.1 Licenses

None

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Closed-loop power control for the PUSCH in dynamic scheduling mode	PUSCH_DYN_SCH_CLOSED_LOOP_SW and PUSCH_INTER_CC_PC_OPT_SW options of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	Inter-CC PUSCH power control optimization takes effect only when both the PUSCH_DYN_SCH_CLOSED_LOOP_SW and PUSCH_INTER_CC_PC_OPT_SW options of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter are selected.
High-frequency TDD	Closed-loop power control for the PUSCH in dynamic scheduling mode	PUSCH_DYN_SCH_CLOSED_LOOP_SW and UL_CA_HYBRID_PC_SW options of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	Hybrid power control for uplink CA takes effect only when both the PUSCH_DYN_SCH_CLOSED_LOOP_SW and UL_CA_HYBRID_PC_SW options of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter are selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Closed-loop power control for the PUSCH in dynamic scheduling mode	PUSCH_DYN_SCH_CLOSED_LOOP_SW option of the NRDUCellUIPcConfig.UlPwrCtrlAlgorithmSwitch parameter	<i>Power Control</i>	PUSCH power control optimization for random access (controlled by the PUSCH_RA_PC_OPTION_SW option of the NRDUCellUIPcConfig.UlPwrCtrlAlgorithmSwitch parameter) can be enabled only when the PUSCH_DYN_SCH_CLOSED_LOOP_SW option of the NRDUCellUIPcConfig.UlPwrCtrlAlgorithmSwitch parameter is selected.

5.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	PUSCH joint power control cannot be enabled in high-speed cells (with NRDUCell.HighSpeedFlag set to HIGH_SPEED).
High-frequency TDD	None	None	None	None

5.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink with independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellCfg.goSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When super uplink with independent SUL deployment is enabled, the NRDUCellCfg.PhrProhibitTimer parameter is ineffective for UEs on which super uplink with independent SUL deployment takes effect.
Low-frequency TDD	PUSCH power control	PUSCH_MIDPOINT_RP_SW option of the NRDUCellCfg.UlPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	When PUSCH joint power control is enabled (by selecting the PUSCH_JOINT_PC_SW option of the NRDUCellCfg.UlPwrCtrlAlgoSwitch parameter), the PUSCH_MIDPOINT_RP_SW option of the NRDUCellCfg.UlPwrCtrlAlgoSwitch parameter must be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Delayed PUSCH power reduction for UEs at a medium distance from the cell center	PUSCH_MIDPOINT_RP_DELAY_SW option of the NRDUCellUPLinkPowerControl parameter	<i>Power Control</i>	Delayed PUSCH power reduction for UEs at a medium distance from the cell center takes effect only when PUSCH power reduction for UEs at a medium distance from the cell center (controlled by the PUSCH_MIDPOINT_RP_SW option of the NRDUCellUPLinkPowerControl parameter) is enabled.
High-frequency TDD	None	None	None	None

5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.2.3.4 Others

None

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 5-7](#) describes the parameters used for optimization.

Table 5-7 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
TPC Value In Msg2	NRDUCellUIPcConfig.TpcValueInMsg2	None	Set this parameter based on the network plan.
PO Nominal PUSCH	NRDUCellUIPcConfig.PoNominalPusch	None	Set this parameter based on the network plan.
Delta Preamble Msg3	NRDUCellUIPcConfig.DeltaPreambleMsg3	None	Set this parameter based on the network plan.
Path Loss Coefficient	NRDUCellUIPcConfig.PathLossCoefficient	None	Set this parameter based on the network plan.
Uplink Power Control Algorithm Switch	NRDUCellUIPcConfig.UlPwrCtrlAlgorithmSwitch	PUSCH_DYN_SCH_CLOSED_LOOP_SW	Select this option.
PUSCH RSRP Threshold	NRDUCellUIPcConfig.PuschRsrpThreshold	None	Set this parameter to its recommended value.
Uplink Power Control Algorithm Switch	NRDUCellUIPcConfig.UlPwrCtrlAlgorithmSwitch	PUSCH_MIDPOINT_RP_SW	Select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Performance Switch	NRDUCellUPLAmc.UPLAmcPerformanceSw	PUSCH_MIDPOINT_RP_DELAY_SW	Select this option.
Uplink Power Control Algorithm Switch	NRDUCellUPLPcConfig.UPLPwrCtrlAlgorithmSwitch	PUSCH_INTER_CC_PC_OPT_SW	Select this option.
Uplink Power Control Algorithm Switch	NRDUCellUPLPcConfig.UPLPwrCtrlAlgorithmSwitch	UL_CA_HYBRID_PC_SW	Select this option.
Uplink Power Control Algorithm Switch	NRDUCellUPLPcConfig.UPLPwrCtrlAlgorithmSwitch	PUSCH_JOINT_PC_SW	It is recommended that this option be selected for all cells in heavy-load or unbalanced-load scenarios.
PUSCH Joint Power Control Cell IoT Thld	NRDUCellUPLPcConfig.PuschJointPcCellIotThld	None	Set this parameter to its recommended value.
PUSCH Joint Power Control User IoT Threshold	NRDUCellUPLPcConfig.PuschJointPcUserIotThld	None	Set this parameter to its recommended value.
PUSCH Joint Power Control MCS Threshold	NRDUCellUPLPcConfig.PuschJointPcMcsThld	None	Set this parameter to its recommended value.
Power Headroom Report Periodic Timer	NRDUCellUPLPcConfig.PhrPeriodicTimer	None	Set this parameter to its recommended value.
Power Headroom Report Prohibit Timer	NRDUCellUPLPcConfig.PhrProhibitTimer	None	Set this parameter to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
UL Preallocation Power Control Enh Sw	NRDUCellAlgoSwitch.UlPreallocPwrCtrlEnhSw	None	It is recommended that this parameter be set to OFF in low-frequency supplementary uplink (SUL) cells or scenarios where UEs in the overlapping area of sectors use MCS 27 (with 256QAM) or MCS 28 (with 64QAM). In other scenarios, it is recommended that this parameter be set to ON .
Uplink Power Control Algorithm Switch	NRDUCellUIPcnfig.UlPwrCtrlAlgoSwitch	UL_CA_PC_OPT_SW	Select this option.
UL Power Control Algorithm Extension Switch	NRDUCellUIPcnfig.UlPwrCtrlAlgoExtSwitch	MF_ACTUAL_PUSCH_ILPC_SW	Select this option to enable multi-frequency actual-PUSCH-based inner-loop power control.
UL Power Control Algorithm Extension Switch	NRDUCellUIPcnfig.UlPwrCtrlAlgoExtSwitch	ULCA_INTER_CC_PWR_ALLOC_ADJ_SW	Select this option to enable inter-CC PUSCH power allocation adjustment for uplink CA UEs.
Uplink CA Power Spectral Density Coefficient	NRDUCellUIPcnfig.UlCaPsdCoefficient	None	Set this parameter to its recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Power Control Algorithm Switch	NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch	PUSCH_RA_PC_OP T_SW	It is recommended that this option be selected in scenarios with limited uplink coverage and be deselected in other scenarios.
UL Power Control Algorithm Extension Switch	NRDUCellULPcConfig.UlPwrCtrlAlgoExtSwitch	MSG4_PWR_CTRL_CFG_SIMPLIFY_SW	Set this parameter based on the network plan.
Complete Message Power Ramping Step	NRDUCellULPcConfig.CmpMsgPwrRampStep	None	Set this parameter based on the network plan.
UL Scheduling Algorithm Switch	NRDUCellULSch.UlSchAlgoSwitch	MSG5_PWR_CTRL_MCS_SEL_OPT_SW	Set this parameter based on the network plan.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Configuring the power offset carried in Msg2
MOD NRDUCELLULPCONFIG: NrDuCellId=0, TpcValueInMsg2=DB6;
//Configuring the nominal Po value of the PUSCH
MOD NRDUCELLULPCONFIG: NrDuCellId=0, PoNominalPusch=-37;
//Configuring the DeltaPreambleMsg3 parameter
MOD NRDUCELLULPCONFIG: NrDuCellId=0, DeltaPreambleMsg3=4;
//Configuring the path loss compensation factor
MOD NRDUCELLULPCONFIG: NrDuCellId=0, PathLossCoeff=ALPHA08;
//Enabling closed-loop power control for the PUSCH in dynamic scheduling mode
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUSCH_DYN_SCH_CLOSED_LOOP_SW-1;
//Configuring the preset RB-level RSRP threshold
MOD NRDUCELLULPCONFIG: NrDuCellId=0, PuschRsrpThld=-71;
//(Only in low frequency bands) Enabling PUSCH joint power control and setting related thresholds after
```

```

selecting the PUSCH_MIDPOINT_RP_SW and PUSCH_MIDPOINT_RP_DELAY_SW options
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUSCH_MIDPOINT_RP_SW-1;
MOD NRDUCELLULPAMC: NrDuCellId=0, ULAmcPerformanceSw=PUSCH_MIDPOINT_RP_DELAY_SW-1;
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUSCH_JOINT_PC_SW-1,
PuschJointPcCelllotThld=5, PuschJointPcUserlotThld=5, PuschJointPcMcsThld=28;
//Configuring the PhrPeriodicTimer parameter
MOD NRDUCELLULPCONFIG: NrDuCellId=0, PhrPeriodicTimer=SF100;
//Configuring the PhrProhibitTimer parameter
MOD NRDUCELLULPCONFIG: NrDuCellId=0, PhrProhibitTimer=SF100;
//(Only in low frequency bands) Enabling power control enhancement for uplink preallocation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlPreallocPwrCtrlEnhSw=ON;
//(Only in low frequency bands) Enabling power control optimization for uplink CA
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=UL_CA_PC_OPT_SW-1;
//(Only in low frequency bands) Enabling multi-frequency actual-PUSCH-based inner-loop power control
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//(Only in low frequency bands) enabling inter-CC PUSCH power allocation adjustment for uplink CA UEs
MOD NRDUCELLULPCONFIG: NrDuCellId=0,
UlPwrCtrlAlgoExtSwitch=ULCA_INTER_CC_PWR_ALLOC_ADJ_SW-1;
//(Only in low frequency bands) Configuring the PSD coefficient for uplink CA
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlCaPsdCoeff=100;
//(Only in high frequency bands) Enabling inter-CC PUSCH power control optimization
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUSCH_INTER_CC_PC_OPT_SW-1;
//(Only in high frequency bands) Enabling hybrid power control for uplink CA
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=UL_CA_HYBRID_PC_SW-1;
//(Only in low-frequency NR TDD) Enabling PUSCH power control optimization for random access
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUSCH_RA_PC_OPT_SW-1;
//Enabling simplified Msg4 power control configuration
MOD NRDUCELLULPCONFIG: NrDuCellId=0,
UlPwrCtrlAlgoExtSwitch=MSG4_PWR_CTRL_CFG_SIMPLIFY_SW-1;
//Configuring the CmpMsgPwrRampStep parameter
MOD NRDUCELLULPCONFIG: NrDuCellId=0, CmpMsgPwrRampStep=STEP_1;
//Selecting the MSG5_PWR_CTRL_MCS_SEL_OPT_SW option
MOD NRDUCELLULSCH: NrDuCellId=0, UlSchAlgoSwitch=MSG5_PWR_CTRL_MCS_SEL_OPT_SW-1;

```

Deactivation Command Examples

MML commands are not required for deactivating this function.

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

None

5.2.4.3 Network Monitoring

After inter-CC PUSCH power control optimization is enabled in high-frequency NR TDD with uplink CA enabled:

- Observe **Cell Uplink Average Throughput (DU)** to monitor the increase in the average uplink cell throughput.
- Observe **User Uplink Average Throughput (DU)** to monitor the increase in the average uplink UE throughput.

Other functions: None

5.3 PUCCH Power Control

5.3.1 Principles

PUCCH power control allows UEs to adjust their transmit power on the PUCCH based on the TPC commands received from the gNodeB.

PUCCH power control ensures the PUCCH performance and reduces the interference to the PUCCH of neighboring cells.

NOTE

The PUCCH carries acknowledgments (ACKs)/negative acknowledgments (NACKs) for downlink data, channel quality indicators (CQIs), and scheduling requests (SRs). UE throughput is severely affected if the probability of incorrect PUCCH demodulation is excessively high.

PUCCH Power Calculation

When UEs initially access or are handed over to a cell, the gNodeB sets the transmit power for the PUCCH based on $P_{0_NOMINAL_PUCCH}$ and σ_{msg2} specified for the cell. This enables CEUs to successfully access the cell and perform services. The PUCCH transmit power is calculated using the following formula:

$$P_{PUCCH}(i) = \min(P_{CMAX,c}, 10 \times \log_{10}(2^{\mu} \times M_{PUCCH}(i)) + P_{0_PUCCH} + PL + \Delta_{TF}(i) + \Delta_{F_PUCCH}(F) + g(i))$$

After $P_{0_NOMINAL_PUCCH}$, $P_{0_UE_PUCCH}$, and $\Delta_{F_PUCCH}(F)$ are delivered by the gNodeB to the UE through RRC signaling, the UE calculates the PUCCH transmit power according to the preceding formula. [Table 5-8](#) defines the variables involved.

Table 5-8 Definitions of the variables in the formula for calculating the PUCCH transmit power

Variable	Definition
i	Indicates the uplink slot number.
$P_{CMAX,c}$	Indicates the maximum transmit power of the UE in cell c .
μ	Indicates the subcarrier spacing configuration factor. The subcarrier spacing is $2^{\mu} \cdot 15\text{kHz}$.
$M_{PUCCH}(i)$	Indicates the number of RBs used by the PUCCH in uplink slot i .

Variable	Definition
P_{o_PUCCH}	Indicates the received PSD expected by the gNodeB. It is calculated using the following formula: $P_{o_PUCCH} = P_{o_NOMINAL_PUCCH} + P_{o_UE_PUCCH},$ where $P_{o_NOMINAL_PUCCH}$ is the target PSD expected by the gNodeB and is specified by the NRDUCellUIPcConfig.PoNominalPucch parameter. $P_{o_UE_PUCCH}$ is the power offset of the UE relative to $P_{o_NOMINAL_PUCCH}$. This value reflects the impact of the UE class, service type, and channel quality on the PUCCH transmit power.
PL	Indicates the downlink path loss estimated by the UE. This value is obtained based on the measured RSRP value and SSB transmit power.
$\Delta_{TF}(i)$	Indicates the impact on the power caused by the number of CQI bits and the number of HARQ signaling bits on the PUCCH. It is determined by the PUCCH format.
$\Delta_{F_PUCCH}(F)$	Reflects the impact of PUCCH formats on the PUCCH transmit power.
$g(i)$	Indicates the adjustment amount for the PUCCH transmit power of the UE. This value is calculated based on $g(i-1)$ and the TPC information sent over the PDCCH. The initial $g(i)$ is calculated using the following formula: $g(0) = \Delta P_{\text{rampup}} + \sigma_{\text{msg2}},$

PUCCH Transmit Power Adjustment

PUCCH transmit power adjustment is introduced to improve PUCCH demodulation performance. This function is controlled by the **PUCCH_INNER_LOOP_SW** option of the **NRDUCellUIPcConfig.UIPwrCtrlAlgoSwitch** parameter. When this option is selected, the gNodeB periodically adjusts the PUCCH transmit power of a UE to adapt to changes in the PUCCH channel environment and enhance PUCCH demodulation performance. The adjustment is based on the difference between the measured SINR and the target SINR on the PUCCH. If the **PUCCH_OUTER_LOOP_SW** option of the **NRDUCellUIPcConfig.UIPwrCtrlAlgoSwitch** parameter is also selected, the gNodeB also dynamically adjusts the target SINR based on the BER and BLER on the PUCCH.

More information on PUCCH transmit power adjustment:

- For the specific adjustment amount

The adjustment amount of the PUCCH transmit power corresponds to $g(i)$ in the formula for calculating the PUCCH transmit power. When the

PUCCH_INNER_LOOP_SW option is deselected, the gNodeB does not adjust the PUCCH transmit power or deliver TPC commands to the UE. In this case, the value of $g(i)$ remains unchanged.

- For the BER threshold on the PUCCH
A low BER threshold will increase the target SINR for PUCCH outer-loop power control and the UE transmit power, causing inter-UE interference. To address this issue, the configurable BER threshold used in PUCCH Reed-Muller (RM) decoding is introduced. When PUCCH outer-loop power control (controlled by the **PUCCH_OUTER_LOOP_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch** parameter) is enabled, the BER threshold used in PUCCH RM decoding can be specified by the **NRDUCellUIPcConfig.PucchBerTargetThld** parameter to reduce inter-UE interference.
- For configurations related to PUCCH inner-loop power control
Configurations related to PUCCH inner-loop power control depend on the settings of the **PUCCH_PC_ALGO_PARAM_OPT_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch** parameter (controlling PUCCH power control parameter optimization) and the **PUCCH_FORMAT_PWR_DIFF_CFG_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch** parameter (controlling PUCCH format power differentiation), as described in [Table 5-9](#).

Table 5-9 Configurations related to PUCCH inner-loop power control

Switch for PUCCH Format Power Differentiation	Switch for PUCCH Power Control Parameter Optimization	Related Configuration
Off	Off	The upper limit of the target RSRP for PUCCH inner-loop power control and the target SINRs for different PUCCH formats are fixed.
Off	On	The upper limit of the target RSRP for PUCCH inner-loop power control and the target SINRs for different PUCCH formats can be configured using parameters.
On	N/A	The upper limit of the target RSRP for PUCCH inner-loop power control, the target SINRs for different PUCCH formats, and the transmit power offsets for different PUCCH formats can be configured using parameters.

Specifically:

- The upper limit of the target RSRP for PUCCH inner-loop power control is specified by the **NRDUCellULPcConfig.PucchRsrpUpperLimit** parameter.
- The target SINR for format-1 PUCCH power control is specified by the **NRDUCellULPcConfig.PucchFormat1PcSinrTarget** parameter.
- The target SINR for format-3 PUCCH power control is specified by the **NRDUCellULPcConfig.PucchFormat3PcSinrTarget** parameter. This parameter value is used as the initial value for target SINR adjustment if the **PUCCH_OUTER_LOOP_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch** parameter is selected. The gNodeB dynamically adjusts the target SINR of PUCCH format 3 based on the initial value as well as the BER and BLER of PUCCH format 3.
- The target SINR for format-4 PUCCH power control is specified by the **NRDUCellULPcConfig.PucchFormat4PcSinrTarget** parameter. This parameter value is used as the initial value for target SINR adjustment if the **PUCCH_OUTER_LOOP_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch** parameter is selected. The gNodeB dynamically adjusts the target SINR of PUCCH format 4 based on the initial value as well as the BER and BLER of PUCCH format 4.
- The transmit power offset for PUCCH format 1 is specified by the **NRDUCellULPcConfig.DeltaFPucchFormat1** parameter.
- The transmit power offset for PUCCH format 3 is specified by the **NRDUCellULPcConfig.DeltaFPucchFormat3** parameter.
- The transmit power offset for PUCCH format 4 is specified by the **NRDUCellULPcConfig.DeltaFPucchFormat4** parameter.

NOTE

When the **PUCCH_INNER_LOOP_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch** parameter is deselected, UEs will falsely detect TPC commands, resulting in higher than expected PUCCH transmit power. It is recommended that this option be selected.

The PUCCH can use RM and polar encoding schemes.

- When the number of transmitted PUCCH bits is less than or equal to 11, the RM encoding scheme is used, and the gNodeB can measure the BER of the PUCCH.
- When the number of transmitted PUCCH bits is greater than or equal to 12, the polar encoding scheme is used, and the gNodeB can measure the BLER of the PUCCH.

5.3.2 Network Analysis

5.3.2.1 Benefits

PUCCH power control properly adjusts the PUCCH transmit power to ensure PUCCH performance, improve the reliability of uplink control information transmission, and increase the average downlink throughput of UEs and the downlink throughput of cells.

After the configurable BER threshold used in PUCCH RM decoding takes effect, the UE transmit power, inter-UE interference, and UE retransmission proportion decrease, which may increase the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**).

5.3.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

PUCCH power control affects the reliability of uplink control information transmission and ACKs/NACKs, as well as the downlink throughput.

PUCCH power control affects the transmission of measurement reports and the random access procedure, thereby affecting the handover success rate.

PUCCH power control affects the transmission reliability over uplink channels and the service drop rate.

When PUCCH format power differentiation is enabled with enhanced CSI period adaptation:

- The interference at the cell edge decreases and the PUCCH demodulation performance improves. As a result, more CEUs may access the cell, increasing the number of online UEs and the uplink and downlink traffic volumes in the cell, and changing the values of cell-level KPIs (including access-related, handover-related, and service-drop-related indicators). Specifically, the following indicators are affected:
 - Average uplink and downlink UE throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**)
 - Average uplink and downlink cell throughputs (**Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**)
 - Average uplink and downlink MCS indexes for UEs
 - Uplink and downlink CCE allocation failure rates

NOTE

- Downlink CCE allocation failure rate (**DL CCE Req Failed Ratio**) = $100 \times (1 - (\text{N.CCE.DL.AggLvl1Num} + \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl4Num} + \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl16Num}) / \text{N.CCE.DL.AllocReq.Num})$
- Uplink CCE allocation failure rate (**UL CCE Req Failed Ratio**) = $100 \times (1 - (\text{N.CCE.UL.AggLvl1Num} + \text{N.CCE.UL.AggLvl2Num} + \text{N.CCE.UL.AggLvl4Num} + \text{N.CCE.UL.AggLvl8Num} + \text{N.CCE.UL.AggLvl16Num}) / \text{N.CCE.UL.AllocReq.Num})$
- The downlink RLC PDU retransmission rate in AM mode due to poll timer expiration (**N.RLC.DL.AM.TrfPDU.Poll.Retrans.TxPackets/N.RLC.DL.AM.TrfPDU.Retrans.TxPackets**) may increase.
- The PUCCH SINR may fluctuate.
- The PUCCH DTX rate may decrease.
- The downlink RLC PDU retransmission rate in AM mode (**N.RLC.DL.AM.TrfPDU.Retrans.TxPackets/N.RLC.DL.AM.TrfPDU.TxPackets**) may decrease.
- The uplink and downlink CCE aggregation levels may decrease.

When the configurable BER threshold used in PUCCH RM decoding takes effect and the UE transmit power is low, the UE retransmission proportion may increase.

5.3.3 Requirements

5.3.3.1 Licenses

None

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

None

5.3.3.2.2 Mutually Exclusive Functions

None

5.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	PUCCH power control parameter optimization	PUCCH_PC_ALGO_PARAM_OPT_SW option of the NRDUCellUIPccOnfig.UlPwrCtrlAlgoSwitch parameter	None	When PUCCH format power differentiation (controlled by the PUCCH_FORMAT_POWER_DIFF_CFG_SW option of the NRDUCellUIMc.UlAmcPerformanceSw parameter) is enabled, PUCCH power control parameter optimization cannot take effect.

5.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.3.4 Others

None

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 5-10](#) describes the parameters used for optimization.

Table 5-10 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
TPC Value In Msg2	NRDUCellULPcCon-fig.TpcValueInMsg2	Set this parameter based on the network plan.
PO Nominal PUCCH	NRDUCellULPcCon-fig.PoNominalPucch	Set this parameter based on the network plan.
Uplink Power Control Algorithm Switch	NRDUCellULPcCon-fig.UlPwrCtrlAlgoSwitch	Select the PUCCH_INNER_LOOP_SW option. Otherwise, UEs will falsely detect TPC commands, resulting in higher than expected PUCCH transmit power.

Parameter Name	Parameter ID	Setting Notes
Uplink Power Control Algorithm Switch	NRDUCellULPcConfig.ULPwrCtrlAlgoSwitch	Select the PUCCH_OUTER_LOOP_SW option.
Uplink Power Control Algorithm Switch	NRDUCellULPcConfig.ULPwrCtrlAlgoSwitch	Select the PUCCH_PC_ALGO_PARAM_OPT_SW option.
PUCCH RSRP Upper Limit	NRDUCellULPcConfig.PucchRsrpUpperLimit	Set this parameter based on the network plan.
PUCCH Format1 Power Control SINR Target	NRDUCellULPcConfig.PucchFormat1PcSnrTarget	Set this parameter based on the network plan.
PUCCH Format3 Power Control SINR Target	NRDUCellULPcConfig.PucchFormat3PcSnrTarget	Set this parameter based on the network plan.
PUCCH Format4 Power Control SINR Target	NRDUCellULPcConfig.PucchFormat4PcSnrTarget	Set this parameter based on the network plan.
Uplink Power Control Algorithm Switch	NRDUCellULPcConfig.ULPwrCtrlAlgoSwitch	Select the PUCCH_FORMAT_PWR_DIFF_CFG_SW option.
DeltaF for PUCCH Format1	NRDUCellULPcConfig.DeltaFPucchFormat1	Set this parameter based on the network plan.
DeltaF for PUCCH Format3	NRDUCellULPcConfig.DeltaFPucchFormat3	Set this parameter based on the network plan.
DeltaF for PUCCH Format4	NRDUCellULPcConfig.DeltaFPucchFormat4	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
PUCCH RM Decoding Bit Error Threshold	NRDUCellULPcConfig.PucchBerTargetThld	It is recommended that this parameter be set to 74 for non-high-speed cells (with NRDUCell.HighSpeedFlag set to LOW_SPEED). It is recommended that this parameter be set to 490 for high-speed cells (with NRDUCell.HighSpeedFlag set to HIGH_SPEED).

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Configuring the power offset carried in Msg2
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, TpcValueInMsg2=DB6;
//Configuring the nominal Po value of the PUCCH
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, PoNominalPucch=-50;
//Enabling inner-loop power control for the PUCCH
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUCCH_INNER_LOOP_SW-1;
//Enabling outer-loop power control for the PUCCH
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUCCH_OUTER_LOOP_SW-1;
//Enabling PUCCH power control parameter optimization
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PUCCH_PC_ALGO_PARAM_OPT_SW-1;
//Configuring the upper limit of the target RSRP for PUCCH inner-loop power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, PucchRsrpUpperLimit=-115;
//Configuring the target SINR for format-1 PUCCH power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, PucchFormat1PcSinrTarget=30;
//Configuring the target SINR for format-3 PUCCH power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, PucchFormat3PcSinrTarget=30;
//Configuring the target SINR for format-4 PUCCH power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, PucchFormat4PcSinrTarget=30;
//Enabling PUCCH format power differentiation
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=
PUCCH_FORMAT_PWR_DIFF_CFG_SW-1;
//Configuring the offset for PUCCH format 1
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, DeltaFPucchFormat1=0;
//Configuring the offset for PUCCH format 3
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, DeltaFPucchFormat3=13;
//Configuring the offset for PUCCH format 4
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, DeltaFPucchFormat4=13;
//Configuring the BER threshold used in PUCCH RM decoding (using a non-high-speed cell as an example)
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, PucchBerTargetThld=74;
```

Deactivation Command Examples

MML commands are not required for deactivating this function.

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

None

5.3.4.3 Network Monitoring

None

5.4 SRS Power Control

5.4.1 Principles

SRS power control is used for uplink channel estimation and uplink timing, ensuring their accuracy.

SRS power control uses variables and TPC commands for PUSCH power control. The SRS transmit power is determined based on the transmission bandwidth and PUSCH power control variables as described in [SRS Power Calculation](#). For PUSCH power control variables, see [Table 5-3](#).

SRS Power Calculation

The SRS transmit power is calculated using the following formula:

$$P_{\text{SRS},f,c}(i,q_s,l) = \min\{P_{\text{CMAX},f,c}(i), 10 \times \log_{10}(2^\mu \times M_{\text{SRS},f,c}(i)) + P_{0,\text{SRS},f,c}(q_s) + \alpha_{\text{SRS},f,c}(q_s) \times PL_{f,c}(q_s) + h_{f,c}(i,l)\}$$

After $P_{0,\text{SRS}}$ and α_{SRS} are delivered by the gNodeB to the UE through RRC signaling, the UE calculates the SRS transmit power according to the preceding formula. [Table 5-11](#) defines the variables involved.

Table 5-11 Definitions of the variables in the formula for calculating the SRS transmit power

Variable	Definition
$P_{\text{SRS},f,c}$	Indicates the SRS transmit power.
i	Indicates the slot number.

Variable	Definition
f	Indicates the carrier number.
q_s	Indicates the SRS resource set number.
$P_{\text{CMAX},f,c}$	Indicates the maximum transmit power of a UE on carrier f in cell c .
μ	Indicates the subcarrier spacing configuration factor. The subcarrier spacing is $2^\mu \cdot 15\text{kHz}$.
$M_{\text{SRS},f,c}$	Indicates the transmission bandwidth of the SRS on carrier f in cell c .
$P_{o,\text{SRS},f,c}$	Indicates the receive PSD expected by the gNodeB for the SRS on carrier f in cell c. For NR TDD: - When the PO_SRS_ADAPT_SW option of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter is deselected, the system uses the NRDUCellUIPcConfig.PoSrs parameter value as $P_{o,\text{SRS}}$. - When the PO_SRS_ADAPT_SW and LARGE_PACKET_UE_SRS_PO_ADPT_SW options of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter are both selected, SRS Po adaptation^a takes effect for all UEs performing downlink large-packet services. The system uses the NRDUCellUIPcConfig.PoSrs parameter value as the initial $P_{o,\text{SRS}}$ and then performs adaptive adjustment based on channel quality and the target SINR for SRS power control to obtain an optimal $P_{o,\text{SRS}}$. (The target SINR for SRS power control is specified by the NRDUCellUIPcConfig.SrsPcSinnrTarget parameter.) If only the PO_SRS_ADAPT_SW option of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter is selected, this function takes effect only for some UEs performing downlink large-packet services. - When the PO_SRS_ADAPT_SW, LARGE_PACKET_UE_SRS_PO_ADPT_SW, and SMALL_PACKET_UE_SRS_PO_ADPT_SW options of the NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch parameter are all selected, SRS Po adaptation takes effect for all UEs performing downlink services (including large-packet and small-packet services). The system uses the NRDUCellUIPcConfig.PoSrs parameter value as the initial $P_{o,\text{SRS}}$ and then performs adaptive adjustment based on channel quality and the target SINR for SRS power control to obtain an optimal $P_{o,\text{SRS}}$. (The target SINR for SRS power control is specified by NRDUCellUIPcConfig.SrsPcSinnrTarget and NRDUCellUIPcConfig.SmallPktUeSrsPcSinnrTarget for large-packet and small-packet UEs, respectively.)

Variable	Definition
$\alpha_{SRS,f,c}$	Indicates the path loss compensation factor on carrier f in cell c .
$PL_{f,c}$	Indicates the downlink path loss estimated by the UE on carrier f in cell c . This value is obtained based on the measured RSRP value and SSB transmit power.
$h_{f,c}(i,l)$	Indicates the adjustment amount for the PUSCH transmit power of the UE. This value is mapped to the TPC command sent over the PDCCH.
a: SRS Po adaptation is supported only in NR TDD low frequency bands. It is recommended that this function be enabled in low-frequency NR TDD cells with massive MIMO enabled and low-frequency NR TDD 4T or 8T low-speed cells (with NRDUCell.HighSpeedFlag set to LOW_SPEED).	

5.4.2 Network Analysis

5.4.2.1 Benefits

SRS power control properly adjusts the SRS transmit power to improve the accuracy of uplink channel estimation and uplink timing.

5.4.2.2 Impacts

The impacts between this function and other functions are described in [5.4.3.2.3 Function Impacts](#).

SRS power control affects the measurement precision of the uplink time alignment (TA) and SINR and the performance of adaptive modulation and coding (AMC), thereby affecting the network throughput.

5.4.3 Requirements

5.4.3.1 Licenses

None

5.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.3.2.1 Prerequisite Functions

None

5.4.3.2.2 Mutually Exclusive Functions

None

5.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink differentiated power control	NRDUCellsRs.UrlcPoOffset	<i>URLLC</i>	When the NRDUCellsRs.UrlcPoOffset parameter controlling uplink differentiated power control is set to a value other than 0 , SRS Po adaptation does not take effect.
High-frequency TDD	None	None	None	None

5.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.4.3.4 Others

None

5.4.4 Operation and Maintenance

5.4.4.1 Data Configuration

5.4.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 5-12](#) describes the parameters used for optimization.

Table 5-12 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
SRS PO	NRDUCellUIPcConfig.PoSrs	None	Set this parameter based on the network plan.
Uplink Power Control Algorithm Switch	NRDUCellUIPcConfig.UlPwrCtrlAlgoSwitch	PO_SRS_ADAPT_SW	You are advised to select this option when SRS interference is weak or the data rates of UEs performing full-buffer services need to be increased. However, if this option is selected in scenarios where downlink full-buffer services are performed at the cell center, the SRS transmit power may decrease, which in turn decreases the average downlink cell throughput.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Power Control Algorithm Switch	NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch	LARGE_PACKET_UE_SRS_PO_ADPT_SW	You are advised to select this option when SRS interference is weak.
Uplink Power Control Algorithm Switch	NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch	SMALL_PACKET_UE_SRS_PO_ADPT_SW	You are advised to select this option when SRS interference is weak.
SRS Power Control SINR Target	NRDUCellULPcConfig.SrsPcSinrTarget	None	Set this parameter based on the network plan.
Small-Packet UE SRS Pwr Ctrl SINR Target	NRDUCellULPcConfig.SmallPktUeSrsPcSinrTarget	None	Set this parameter based on the network plan.

5.4.4.1.2 Using MML Commands

Before using MML commands, refer to [5.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Configuring the SRS Po value
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, PoSrs=-37;
//Turning on the SRS Po adaptation switch (recommended when SRS interference is weak or the data rates
of UEs performing full-buffer services need to be increased)
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoSwitch=PO_SRS_ADAPT_SW-1;
//Turning on the SRS Po adaptation switch for common large-packet UEs (recommended when SRS
interference is weak)
MOD NRDUCELLULPCCONFIG: NrDuCellId=0,
UlPwrCtrlAlgoSwitch=LARGE_PACKET_UE_SRS_PO_ADPT_SW-1;
//Turning on the SRS Po adaptation switch for small-packet UEs (recommended when SRS interference is
weak)
MOD NRDUCELLULPCCONFIG: NrDuCellId=0,
UlPwrCtrlAlgoSwitch=SMALL_PACKET_UE_SRS_PO_ADPT_SW-1;
//Configuring the target SINR for SRS power control for large-packet UEs
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, SrsPcSinrTarget=100;
//Configuring the target SINR for SRS power control for small-packet UEs
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, SmallPktUeSrsPcSinrTarget=80;
```

Deactivation Command Examples

MML commands are not required for deactivating this function.

5.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.4.2 Activation Verification

None

5.4.4.3 Network Monitoring

None

5.5 UE Transmit Power Control

5.5.1 Principles

3GPP specifications impose restrictions and requirements on the indicators related to UE transmit power, including maximum output power, maximum power reduction (MPR), additional maximum power reduction (A-MPR), and additional spurious emissions. For details, see chapter 6 "Transmitter characteristics" in 3GPP TS 38.101-2 V15.8.0.

In high-frequency NR TDD, UE transmit power control enables the base station to control the A-MPR and additional spurious emissions of UEs, reducing additional spectrum interference. The UE determines the A-MPR and additional spurious emissions based on the network signaling (NS) value mapped to the *additionalSpectrumEmission* IE value. The *additionalSpectrumEmission* IE is delivered by the gNodeB to the UE through RRC signaling messages or system information blocks (SIBs). [Table 5-13](#) describes the mapping between NS values and *additionalSpectrumEmission* IE values. For details, see sections 6.2.3 "UE maximum output power with additional requirements" and 6.5.3.2 "Additional spurious emissions" in 3GPP TS 38.101-2 V15.8.0.

Table 5-13 NS value mapping table

NR Band	Value of <i>additionalSpectrumEmission</i>							
	0	1	2	3	4	5	6	7
n257	NS_200	-	-	-	-	-	-	-
n258	NS_200	NS_201	-	-	-	-	-	-

n260	NS_20 0	-	-	-	-	-	-	-
n261	NS_20 0	-	-	-	-	-	-	-

- If the **ADDL_SPCT_EMSN1_CONFIG_SW** option of the **gNBFreqBandConfig.AddlSpectEmsnConfigSwitch** parameter is selected, the gNodeB delivers the *additionalSpectrumEmission* IE with the value 1 to the UE.
- If the **ADDL_SPCT_EMSN1_CONFIG_SW** option of the **gNBFreqBandConfig.AddlSpectEmsnConfigSwitch** parameter is deselected, the gNodeB does not deliver the *additionalSpectrumEmission* IE with the value 1 but delivers the IE with the default value 0 according to the 3GPP specifications.

In low frequency bands, the *p-Max* IE in SIB1 is used to limit the maximum UE transmit power according to 3GPP TS 38.331. UE transmit power control in low frequency bands is controlled by the **NRDUCellSelConfig.MaxUeTransmitPower** parameter specifying the maximum UE transmit power.

- If the **NRDUCellSelConfig.MaxUeTransmitPower** parameter is set to **254**, SIB1 does not contain the *p-Max* IE. In this case, the maximum UE transmit power is the maximum output power corresponding to the UE power class specified in section 6.2.1 "UE maximum output power" in 3GPP TS 38.101-1 V17.7.0.
- If the **NRDUCellSelConfig.MaxUeTransmitPower** parameter is set to **255**, SIB1 contains the *p-Max* IE.
In low-frequency NR TDD, the maximum UE transmit power is 26 dBm for bands n41, n77, n78, and n79 and 23 dBm for other bands.
- If the **NRDUCellSelConfig.MaxUeTransmitPower** parameter is set to an integer ranging from **-30** to **33**, the value of the *p-Max* IE in SIB1 is equal to the value of this parameter. That is, the maximum UE transmit power (*p-Max*) is restricted on the base station side. The smaller the value of this parameter, the lower the maximum UE transmit power. The larger the value of this parameter, the higher the maximum UE transmit power. However, the maximum UE transmit power cannot exceed the maximum output power corresponding to the UE power class.

5.5.2 Network Analysis

5.5.2.1 Benefits

In high-frequency NR TDD, UE transmit power control enables the base station to control the A-MPR and additional spurious emissions of UEs by configuring the *additionalSpectrumEmission* value, thereby reducing additional spectrum interference.

UE transmit power control in low frequency bands is controlled by the **NRDUCellSelConfig.MaxUeTransmitPower** parameter specifying the maximum UE transmit power to meet requirements in different scenarios.

5.5.2.2 Impacts

The impacts between this function and other functions are described in [5.5.3.2.3 Function Impacts](#).

In high-frequency NR TDD, when the **ADDL_SPCT_EMSN1_CONFIG_SW** option of the **gNBFreqBandConfig.AddlSpctEmsnConfigSwitch** parameter is selected, the maximum UE transmit power may decrease, resulting in a decrease in the uplink traffic volume; when this option is deselected, network performance is not affected.

In low frequency bands, the smaller the **NRDUCellSelConfig.MaxUeTransmitPower** parameter value, the lower the maximum UE uplink transmit power.

5.5.3 Requirements

5.5.3.1 Licenses

None

5.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.5.3.2.1 Prerequisite Functions

None

5.5.3.2.2 Mutually Exclusive Functions

None

5.5.3.2.3 Function Impacts

None

5.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

In high-frequency NR TDD: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

In low frequency bands:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

In high-frequency NR TDD:

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

In low frequency bands: This function does not depend on RF modules.

5.5.3.4 Others

None

5.5.4 Operation and Maintenance

5.5.4.1 Data Configuration

5.5.4.1.1 Data Preparation

Table 5-14 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 5-14 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Addl Spct Emission Config Switch	gNBFreqBandConfig.AddlSpctEmsnConfigSwitch	ADDL_SPCT_EMSN1_CONFIG_SW	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Max UE Transmit Power	NRDUCellSel Config. <i>MaxUeTransmitPower</i>	None	Set this parameter based on the network plan.

5.5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//(Only in high frequency bands) Enabling UE transmit power control  
MOD GNBFREQBANDCONFIG: FrequencyBand=N258, AddSpctEmsnConfigSwitch  
=ADDL_SPCT_EMSN1_CONFIG_SW-1;  
//(Only in low frequency bands) Enabling UE transmit power control (using -9 dBm as an example)  
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MaxUeTransmitPower=-9;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//(Only in high frequency bands) Disabling UE transmit power control  
MOD GNBFREQBANDCONFIG: FrequencyBand=N258, AddSpctEmsnConfigSwitch  
=ADDL_SPCT_EMSN1_CONFIG_SW-0;  
//(Only in low frequency bands) Disabling UE transmit power control  
MOD NRDUCELLSELCONFIG: NrDuCellId=0, MaxUeTransmitPower=255;
```

5.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.4.2 Activation Verification

The activation verification procedure for high-frequency NR TDD is as follows:

- Step 1** Start Uu signaling tracing on the MAE-Access as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the

displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

- Step 2** Enable a UE to access the network and observe the RRC_CONN_RECFG message to check whether this function takes effect: Check the fields in the *CellGroupConfig* IE in the RRC_CONN_RECFG message. If the value of *additionalSpectrumEmission* is 1, this function has taken effect.

----End

The activation verification procedure for low frequency bands is as follows:

- Step 1** Start Uu signaling tracing on the MAE-Access as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

- Step 2** Check the value of the *p-Max* IE in SIB1 to verify whether this function has taken effect.

----End

5.5.4.3 Network Monitoring

None

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Downlink Scheduling*
- *High Speed Mobility*
- *Super Uplink*
- *CoMP*
- *NR Inter-Carrier Dynamic Power Allocation*
- *LTE and NR Power Allocation*
- *URLLC*
- *Fusion Cell (Low-Frequency TDD)*
- *UE Power Saving*
- *RedCap*
- *Modulation Schemes*
- *mmWave Super Coverage (High-Frequency TDD)*